

A global spectral gap for Metropolis-adjusted Langevin algorithm with a uniformly randomized step size

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Abstract

Let $\pi(dx) \propto e^{-U(x)} dx$ on $\mathbb{R}^d$, where U is continuously differentiable and m -strongly convex with a globally L -Lipschitz gradient, $0 < m \leq L < \infty$, and $\kappa = L/m$. It is known that, under warm-start assumptions, fixed-step Metropolis-adjusted Langevin algorithm (MALA) with properly tuned step size has mixing time of order $\kappa\sqrt{d}$ up to logarithmic factors. By contrast, when the condition number is bounded away from one, no single fixed step size yields a matching spectral-gap lower bound of order $(\kappa\sqrt{d})^{-1}$ uniformly over this target class. We establish a global spectral-gap lower bound for MALA with a uniformly randomized step size. At each iteration, the algorithm draws h uniformly from $(0, H)$ and performs one ordinary MALA transition. Choosing H of order

$$\frac{1}{L\sqrt{d[1 + \log(d+1) + \log \kappa]}}$$

yields a right spectral-gap lower bound of order

$$\frac{1}{\kappa\sqrt{d[1 + \log(d+1) + \log \kappa]}},$$

uniformly over the target class. A key ingredient in the proof is a Cheeger-type inequality for aggregating estimates of the one-step flow of MALA out of measurable sets at various step-size scales. It allows the scale used to control the flow to depend on the set and avoids the additional loss that would result from first estimating the conductance and then applying the standard Cheeger inequality. This work was developed with substantial assistance from ChatGPT, which suggested the uniformly randomized-step approach, developed the principal proof arguments, and generated the simulation and Lean 4 code. The human author checked and verified the mathematical content and takes full responsibility for the results.

1 Introduction

For a target probability distribution π on $\mathbb{R}^d$ with density proportional to e^{-U} , the Metropolis-adjusted Langevin algorithm (MALA) proposes

$$Y = x - h\nabla U(x) + \sqrt{2h} Z, \text{ where } Z \sim N_d(0, I_d),$$

and applies the Metropolis–Hastings correction. MALA is one of the most fundamental Markov chain Monte Carlo (MCMC) algorithms and has been studied extensively. Its classical geometric properties and high-dimensional scaling were developed by [Roberts and Tweedie \[1996\]](#), [Roberts and Rosenthal \[1998\]](#); it is also closely related to Hamiltonian Monte Carlo [[Neal, 2011](#), [Wu et al., 2022](#), [Chen and Gatmiry, 2023](#)].

For an m -strongly convex and L -smooth potential $U(\cdot)$, [Wu et al. \[2022\]](#), strengthening results in [Chewi et al. \[2021\]](#), show that the mixing time of (lazy) MALA from a warm start has the minimax $\kappa\sqrt{d}$ dependence, up to logarithmic factors, where $\kappa = L/m$ is the condition number. This rate is achieved with a MALA step size h of order $(L\sqrt{d})^{-1}$ up to logarithmic factors that depend on the desired accuracy. The proofs in [Chewi et al. \[2021\]](#) and [Wu et al. \[2022\]](#) control acceptance on a high-probability region and then use an s -conductance argument [[Lovász and Simonovits, 1993](#)]. They therefore do not lower-bound the ordinary conductance over sets of every positive stationary mass, and their arguments do not yield a matching global Poincaré bound of the form

$$\text{Gap}_R(P_h) \gtrsim \frac{1}{\kappa\sqrt{d}}, \quad (1)$$

where P_h is the Markov kernel of MALA with step size h , $\text{Gap}_R(\cdot)$ denotes the right spectral gap, and $\gtrsim$ suppresses universal constants and the logarithmic factors. (Recall that, for a lazy reversible Markov chain, the mixing time is roughly inversely proportional to the right spectral gap, so a gap of order $(\kappa\sqrt{d})^{-1}$ implies mixing time of order $\kappa\sqrt{d}$.) Under only strong convexity and gradient Lipschitzness, a fixed-step global Poincaré bound at the $d^{-1/2}$ scale is impossible uniformly over the target class. Related fixed-step spectral-gap obstructions were established by [Chewi et al. \[2021\]](#), [Lee et al. \[2021\]](#), [Wu et al. \[2022\]](#). In particular, [Lee et al. \[2021, Theorem 2\]](#) give a spectral gap upper bound of order $\log d/(\kappa d)$ when $\kappa = O(d^4)$. In Proposition 2.3, it is shown that, when $\kappa \geq \kappa_0 > 1$ for some constant κ_0 , for any $h > 0$, there exists an m -strongly convex and L -smooth potential U such that $\text{Gap}_R(P_h)$ is upper bounded by a κ_0 -dependent multiple of the maximum of $\log(\kappa d)/(\kappa d)$ and e^{-cd} , where $c > 0$ is universal.

A global spectral-gap bound is useful for more than deriving a mixing-time estimate. Let P be a lazy (or positive semi-definite), π -reversible Markov kernel. If μ is an initial distribution with $d\mu/d\pi \in L^2(\pi)$, then

$$\|\mu P^n - \pi\|_{\text{TV}} \leq \frac{1}{2} \left\| \frac{d\mu}{d\pi} - 1 \right\|_{L^2(\pi)} [1 - \text{Gap}_R(P)]^n. \quad (2)$$

Thus a spectral gap certifies convergence all the way to stationarity, rather than only for a prescribed accuracy [[Kontoyiannis and Meyn, 2012](#), [Roberts and Rosenthal, 1997](#), [Rudolf, 2012](#)]. More importantly for Monte Carlo, for a stationary reversible chain and a function in $L^2(\pi)$, a positive right spectral gap implies a central limit theorem for Markov-chain averages and controls their asymptotic variance [[Kipnis and Varadhan, 1986](#)]. Related spectral arguments also yield finite-sample variance, mean-square error, and concentration bounds for Markov-chain averages [[Rudolf, 2012](#), [Paulin, 2015](#)]. By contrast, a favorable warm-start mixing bound can coexist with very high autocorrelation for certain $L^2(\pi)$ test functions.

Quantitative global spectral-gap results for MALA are considerably sparser than mixing-time bounds. The classical analysis of [Roberts and Tweedie \[1996\]](#) gives qualitative criteria for geometric ergodicity, rather than dimension-explicit lower bounds on the gap, while [Bou-Rabee and Hairer \[2013\]](#) show that, for drifts that are not globally Lipschitz, MALA may fail to have a spectral gap even when the underlying Langevin diffusion does. [Wu et al. \[2022\]](#) use spectral-gap upper bounds of order mh for specially constructed targets to establish their minimax mixing-time lower bound, but the existing fixed-step analyses do not provide a matching global spectral-gap lower bound when h is of order $(L\sqrt{d})^{-1}$. More recently, [Liu and Tong \[2026\]](#) establish $\text{Gap}_R(P_h) \gtrsim (\kappa^2 d)^{-1}$ when h is of order $(\kappa L d)^{-1}$. This spectral-gap bound gives a mixing-time bound of order $\kappa^2 d$, which is significantly worse than the $\kappa\sqrt{d}$ dependence achieved with a step size of order $(L\sqrt{d})^{-1}$.

We study the randomized kernel

$$\bar{P}_H = \frac{1}{H} \int_0^H P_h \, dh.$$

Operationally, the algorithm samples $h \sim \text{Unif}(0, H)$ and then performs one ordinary MALA step. It uses one proposal and has essentially the same computational cost as fixed-step MALA. By linearity of detailed balance, it is exactly π -reversible. Under the stated strong-convexity and gradient-Lipschitz assumptions, [Theorem 2.1](#) gives a global spectral-gap lower bound for every $H > 0$. In particular, [Corollary 2.2](#) shows that choosing H of order $1/[L\sqrt{d}[1 + \log(d+1) + \log \kappa]]$ gives a gap lower bound of order

$$\frac{1}{\kappa\sqrt{d[1 + \log(d+1) + \log \kappa]}}.$$

Thus the $(\kappa\sqrt{d})^{-1}$ dependence is achieved with only a square-root logarithmic loss. The same corollary retains the bound for the larger endpoint H of order $(L\sqrt{d})^{-1}$, with a full logarithmic loss.

With the smaller endpoint H , the resulting mixing-time upper bound for the half-lazy kernel $(I + \bar{P}_H)/2$ has $\kappa\sqrt{d[1 + \log(d+1) + \log \kappa]} \log(M/\epsilon)$ dependence, where ϵ is the desired accuracy and M is a warmness parameter for the initial distribution. The scale parameter H does not need to depend on ϵ . This contrasts the mixing time bounds for fixed-step MALA in [Chewi et al. \[2021\]](#), [Wu et al. \[2022\]](#), where the step size h is dependent on a prescribed ϵ .

Randomization of Metropolis tuning parameters has been advocated as a way to improve robustness. In the setting most directly relevant here, [Grazzi et al. \[2026\]](#) study the randomized kernel $\int P_h \nu(dh)$, where ν is a distribution on $(0, \infty)$. They show that randomized kernels inherit spectral gaps and weak Poincaré inequalities from suitable fixed-step components and that randomization can turn severe sensitivity to step-size misspecification into a much milder deterioration of the spectral gap. The intuition is that a randomized method may sacrifice some peak asymptotic efficiency in exchange for performing reasonably over a much wider range of scales. Related benefits of randomization are known for Hamiltonian methods [[Bou-Rabee and Sanz-Serna, 2017](#), [Apers et al., 2024](#)].

As shown in [Grazzi et al. \[2026\]](#), the randomized kernel satisfies the inheritance bound $\text{Gap}_R(\bar{P}_H) \geq H^{-1} \int_0^H \text{Gap}_R(P_h) \, dh$. This inequality alone, however, does not yield the $(\kappa\sqrt{d})^{-1}$ rate sought here, because no available fixed-step result establishes $\text{Gap}_R(P_h) \gtrsim (\kappa\sqrt{d})^{-1}$, regardless of h . To be more specific, existing conductance lower bounds, which are translated into spectral-gap lower bounds via Cheeger’s inequality [[Lawler and Sokal, 1988](#)], are considerably smaller than desired at every scale. Recall that Cheeger’s inequality asserts that $\text{Gap}_R(P_h)$ is lower bounded by $\phi(P_h)^2/2$, where $\phi(P_h)$ is the global conductance:

$$\phi(P_h) := \inf_{S: 0 < \pi(S) \leq 1/2} \frac{\mathcal{J}_{P_h}(S, S^c)}{\pi(S)}, \quad \mathcal{J}_{P_h}(S, S^c) = \int_S P_h(x, S^c) \pi(dx).$$

To apply Cheeger’s inequality, we want a lower bound on $\phi(P_h)$ that is of order $(\kappa\sqrt{d})^{-1/2}$ for a sufficiently large set of $h \in [0, H]$. [Liu and Tong \[2026\]](#) obtain a lower bound on $\phi(P_h)$ of order $(\kappa^2 d)^{-1/2}$ for $h \lesssim (\kappa L d)^{-1}$. For h of order $(L\sqrt{d})^{-1}$, [Wu et al. \[2022\]](#) obtain a lower bound of order $(\kappa\sqrt{d})^{-1/2}$ on $\mathcal{J}_{P_h}(S, S^c)/\pi(S)$ when $\pi(S)$ exceeds a certain threshold; below that threshold, however, the ratio can scale very poorly with d .

Our argument exploits information that the preceding inheritance bound discards: the step-size range used to control the flow out of a set may depend on the stationary mass of that set. For each measurable $S \subset \mathbb{R}^d$, we select a range of step sizes tailored to $\pi(S)$, using progressively smaller

steps as $\pi(S)$ decreases. When $\pi(S)$ is not too small, comparatively larger steps can be used: the Metropolis acceptance rate is lower bounded by a positive constant over S , outside an exceptional set with small stationary mass [Chewi et al., 2021, Wu et al., 2022]. Writing $\mathcal{K}_t = (2/t) \int_{t/2}^t P_h \, dh$ for the kernel averaged over the upper dyadic half of $(0, t)$, one can then lower-bound the one-step flow $\mathcal{J}_{\mathcal{K}_t}(S, S^c)$ by a suitable multiple of $\pi(S)$. For smaller sets, the exceptional region in the preceding argument can no longer be ignored, and we instead use a smaller, globally safe step size which leads to a constant uniform lower bound on the acceptance rate over the whole space. Gaussian isoperimetry for strongly log-concave distributions then supplies the corresponding lower bound on the flow out of S .

Finally, we combine the flow estimates obtained at the different scales using a Cheeger-type aggregation lemma (Lemma 3.5) for mixtures of reversible Markov kernels. The lemma lets us use relatively large steps for sets of moderate stationary mass and smaller steps for rarer sets, without losing an additional factor that would result from first estimating the conductance and then applying the standard Cheeger inequality. The closest analytic predecessor is the general Cheeger inequality of Chen and Wang [2000], which can be used to derive Lemma 3.5. We give a direct proof and a more transparent formulation (Theorem 3.6) suitable for the present setting.

An accompanying Lean 4 development provides proof scripts for the main theorem and other principal theorem-level conclusions, including Theorems 2.1 and 3.6, Proposition 2.3, and Lemma 3.5. The scope of these scripts and the verification status of the revised first-order interface are described in Section 5.

The rest of the article is organized as follows. The main result is stated in Section 2. Section 3 lays out the ingredients of the proof of the main theorem. The proof is completed in Section 4. The description for Lean 4 formalization in Section 5 is followed by a simulation experiment comparing fixed- and randomized-step MALA in Section 6. The Appendices contain technical proofs and lemmas.

Relation to existing work

Controlling the rejection rate is central to the analysis of MALA. There are several approaches to controlling MALA rejection at the $d^{-1/2}$ step-size scale. Chewi et al. [2021] use a projection characterization of Metropolis adjustment and compare the Euler proposal with exact Langevin diffusion. Wu et al. [2022] instead compare one leapfrog step with continuous Hamiltonian dynamics. Earlier quantitative rejection estimates and Wasserstein convergence bounds for Langevin-based Metropolis algorithms targeting perturbations of Gaussian measures were obtained by Eberle [2014] under higher-order regularity assumptions. Mangoubi and Vishnoi [2019] analyze MALA acceptance through Hamiltonian energy-conservation error under third- and fourth-order regularity conditions, including for nonconvex targets. Earlier pathwise comparisons between Metropolized integrators and their limiting diffusions include Bou-Rabee and Vanden-Eijnden [2010], Bou-Rabee and Hairer [2013].

Our appendix follows the diffusion-comparison philosophy. The stationary increment estimate follows from the Lyons–Zheng forward–backward decomposition [Lyons and Zheng, 1988, Section 1, especially equation (1.7)]; see also Fukushima et al. [2011, Theorem 5.7.1]. We give an elementary proof for Langevin diffusion by writing the stochastic differential equation along the forward and reversed paths and subtracting the resulting expressions to cancel the drift. A Girsanov change of path law then compares the exact diffusion with the drift frozen at its initial state. Endpoint conditioning and the Metropolis–Hastings meet identity [Tierney, 1998, Billera and Diaconis, 2001] convert the path estimate into an $L^p(\pi)$ rejection bound.

The close-coupling and isoperimetric part of the argument is standard in modern Metropolis

analyses [Dwivedi et al., 2019, Wu et al., 2022, Andrieu et al., 2024, Liu and Tong, 2026]. The use of a scale indexed by set size is related in spirit to average conductance, evolving sets, blocking conductance, and spectral profiles [Lovász and Kannan, 1999, Morris and Peres, 2005, Goel et al., 2006, Kannan et al., 2006].

The closest analytic predecessor of our aggregation lemma is Chen and Wang [2000]. They prove a general version of Cheeger’s inequality that applies in settings considerably broader than the Markov-kernel setting considered here and extends the earlier results of Lawler and Sokal [1988]. Their result can be used to derive our aggregation lemma, although the proof we give is direct and self-contained. The part specific to our argument is the way the lemma is combined with the set-dependent choice of scale: different sets may require different step-size ranges. We use this idea to combine flow estimates over a geometric sequence of step-size ranges for randomized-step MALA.

A complementary line of research develops alternative gradient-based samplers and studies the number of oracle queries needed to reach a prescribed distributional accuracy. Examples include randomized-midpoint discretizations of underdamped Langevin dynamics [Shen and Lee, 2019] and the path-space rejection methods of Chen et al. [2026]. Under their regularity and initialization assumptions, the latter achieve expected gradient-query complexity with the more favorable $\kappa^{2/3}d^{1/3}$ dependence, up to logarithmic factors, for sampling in Rényi divergence. Like the mixing time bounds in Chewi et al. [2021], Wu et al. [2022], these results concern a prescribed level of accuracy, on which the parameters in the sampling algorithm depend. As mentioned earlier, the global spectral-gap bound herein supplies more than a prescribed-accuracy sampling guarantee. It provides guarantees for both convergence and long-run Monte Carlo estimation, and it does so in a way that the Markov kernel can stay fixed for all levels of accuracy.

Generative AI disclosure

This project began when the author asked ChatGPT 5.6 Sol (OpenAI) whether MALA admits a useful global spectral-gap bound at the $d^{-1/2}$ step-size scale. ChatGPT suggested studying a uniformly randomized-step version of MALA—an algorithmic variant that had previously appeared in the literature—and developed the principal proof strategy and initial versions of the major mathematical arguments in this article. It also assisted in drafting and revising the manuscript and generated all code used for the numerical experiments and the Lean 4 formal verification. The human author checked the relevant literature, scrutinized and revised the AI-generated arguments, verified the mathematical claims and computations, inspected the simulation methodology and results, and confirmed that the Lean development was kernel-checked without proof placeholders or problem-specific axioms. The human author takes full responsibility for the content and correctness of the article.

2 Setup and Main Result

Assume throughout, except where explicitly stated otherwise, that d is a positive integer and that $U : \mathbb{R}^d \rightarrow \mathbb{R}$ is continuously differentiable, m -strongly convex, and has a globally L -Lipschitz gradient, where $0 < m \leq L < \infty$. Explicitly, for every $x, y \in \mathbb{R}^d$,

$$\begin{aligned} U(y) &\geq U(x) + \langle \nabla U(x), y - x \rangle + \frac{m}{2} \|y - x\|^2, \\ \|\nabla U(y) - \nabla U(x)\| &\leq L \|y - x\|. \end{aligned} \tag{3}$$

Strong convexity implies that e^{-U} is integrable. Let $\kappa := L/m \geq 1$, and define the target distribution

$$\pi(\mathrm{d}x) = \frac{e^{-U(x)}}{\int_{\mathbb{R}^d} e^{-U(u)} \mathrm{d}u} \mathrm{d}x.$$

For $h > 0$, let $Q_h(x, \cdot)$ be the Gaussian law

$$N_d(x - h\nabla U(x), 2hI_d).$$

The fixed-step MALA kernel is

$$P_h(x, \mathrm{d}y) = \alpha_h(x, y) Q_h(x, \mathrm{d}y) + \left[1 - \int \alpha_h(x, z) Q_h(x, \mathrm{d}z)\right] \delta_x(\mathrm{d}y),$$

where

$$\alpha_h(x, y) = 1 \wedge \frac{e^{-U(y)} q_h(y, x)}{e^{-U(x)} q_h(x, y)}$$

and q_h is the density of Q_h . The usual Metropolis–Hastings argument makes P_h π -reversible [Tierney, 1998]; consequently, so is the randomized-step MALA kernel

$$\overline{P}_H(x, \mathrm{d}y) = \frac{1}{H} \int_0^H P_h(x, \mathrm{d}y) \mathrm{d}h.$$

For probability measures μ, ν on a measurable space $(\Omega, \mathcal{F})$, write

$$\mathrm{TV}(\mu, \nu) = \sup_{A \in \mathcal{F}} |\mu(A) - \nu(A)|.$$

For a probability measure ϖ , let $L^2(\varpi)$ be the space of measurable functions $f : \Omega \rightarrow \mathbb{R}$ such that $\int f^2 \mathrm{d}\varpi < \infty$. For a ϖ -reversible kernel P and $f \in L^2(\varpi)$, define the Dirichlet form

$$\mathcal{E}_P(f, f) = \frac{1}{2} \int [f(x) - f(y)]^2 \varpi(\mathrm{d}x) P(x, \mathrm{d}y).$$

The right, or Poincaré, spectral gap is defined as

$$\mathrm{Gap}_R(P) = \inf_{\substack{f \in L^2(\varpi) \\ \mathrm{Var}_\varpi(f) > 0}} \frac{\mathcal{E}_P(f, f)}{\mathrm{Var}_\varpi(f)}.$$

This quantity lies in $[0, 2]$, and controls the convergence properties of the Markov chain associated with P if P is lazy or positive semidefinite.

The main result of this paper is stated below. Its proof is given in Sections 3 and 4.

Theorem 2.1. *There are universal constants $A_0 \in [1, \infty)$, $b_0 \in (0, 1/2]$, and $c_0 \in (0, \infty)$ such that the following holds. Define*

$$p_\star = A_0[1 + \log(d+1) + \log \kappa].$$

Then, for every $H \in (0, \infty)$,

$$\mathrm{Gap}_R(\overline{P}_H) \geq c_0 \frac{m}{H} \min \left\{ H, \frac{b_0}{L} \max \left\{ \frac{1}{\sqrt{p_\star(d+p_\star)}}, \frac{1}{d} \right\} \right\}^2. \quad (4)$$

For the lazy kernel $(I + \overline{P}_H)/2$, the right-hand side is divided by two.

Corollary 2.2. Under the notations of Theorem 2.1, let $c \in (0, \infty)$, and let $H = c/(L\sqrt{d})$. Then

$$\text{Gap}_R(\bar{P}_H) \geq \frac{c_0}{\kappa\sqrt{d}} \min \left\{ c, \frac{b_0^2}{c} \max \left\{ \frac{d}{p_*(d+p_*)}, \frac{1}{d} \right\} \right\}.$$

In particular, since $\min\{p_*(d+p_*)/d, d\} \leq 2p_*$,

$$\text{Gap}_R(\bar{P}_H) \geq \frac{c_0}{\kappa\sqrt{d}} \min \left\{ c, \frac{b_0^2}{2cp_*} \right\}.$$

Choosing $c = c'/\sqrt{p_*}$ for some $c' \in (0, \infty)$ gives

$$\text{Gap}_R(\bar{P}_H) \geq \frac{c_0}{\kappa\sqrt{d}p_*} \min \left\{ c', \frac{b_0^2}{2c'} \right\}.$$

This spectral gap bound for randomized-step MALA is in sharp contrast with the following upper bound on the spectral gap of fixed-step MALA, which is proved in Section A.

Proposition 2.3. Fix $\kappa_0 > 1$. There are a universal constant $c \in (0, \infty)$ and a constant $C \in (0, \infty)$ depending only on κ_0 such that the following holds. Let $d \geq 2$, $0 < m < L$, and $\kappa = L/m \geq \kappa_0$. Then

$$\sup_{h>0} \inf_{\substack{U \in C^\infty(\mathbb{R}^d) \\ mI_d \leq \nabla^2 U(x) \leq LI_d, \ x \in \mathbb{R}^d}} \text{Gap}_R(P_{U,h}) \leq C \max \left\{ \frac{\log(\kappa d)}{\kappa d}, e^{-cd} \right\}. \quad (5)$$

Here $P_{U,h}$ denotes the MALA kernel with step size h and target density proportional to e^{-U} .

The proof of Proposition 2.3 utilizes constructions from Chewi et al. [2021], Lee et al. [2021], Wu et al. [2022]. These works construct a hard target distribution and use it to bound the spectral gap of fixed-step MALA from above. Their results do not readily yield the upper bound herein. We use a modified version of the construction in Wu et al. [2022] to obtain the bound.

A mixing-time bound follows from Corollary 2.2. Let $\mu \ll \pi$ be a probability distribution on $\mathbb{R}^d$ with $d\mu/d\pi \in L^2(\pi)$. For a Markov kernel P with stationary distribution π and $\epsilon > 0$, define

$$n_{\text{mix}}(P, \mu, \epsilon) = \inf \{n \in \{0, 1, \dots\} : \|\mu P^n - \pi\|_{\text{TV}} \leq \epsilon\}, \quad \inf \emptyset = \infty.$$

Then the following holds.

Corollary 2.4. Under the notations of Theorem 2.1, let $H = c/(L\sqrt{d}p_*)$, where $c > 0$, and put

$$\ell_{\mu, \epsilon} = \log \max \left\{ 1, \frac{1}{2\epsilon} \left\| \frac{d\mu}{d\pi} - 1 \right\|_{L^2(\pi)} \right\}.$$

Then

$$n_{\text{mix}} \left(\frac{I + \bar{P}_H}{2}, \mu, \epsilon \right) \leq \left\lceil \frac{2\ell_{\mu, \epsilon}}{\text{Gap}_R(\bar{P}_H)} \right\rceil \leq \left\lceil C\ell_{\mu, \epsilon} \kappa\sqrt{dp_*} \right\rceil,$$

where $C \in (0, \infty)$ depends only on c .

Proof. Since $0 < \text{Gap}_R(\bar{P}_H)/2 \leq 1$, the inequality $1 - x \leq e^{-x}$ for $x \in [0, 1]$, together with (2) and Corollary 2.2 gives the desired mixing time bound. The logarithm defining $\ell_{\mu, \epsilon}$ is finite and nonnegative, including when $\mu = \pi$. If $\ell_{\mu, \epsilon} = 0$, the initial total-variation error is already at most ϵ , so the mixing time is zero. In the edge case $\text{Gap}_R(\bar{P}_H)/2 = 1$, (2) shows that the chain has law π after one step from μ ; the same bound remains valid. $\square$

3 Ingredients of the Proof

We now describe the main ingredients of the proof of Theorem 2.1.

For $t > 0$, define the MALA kernel averaged over the upper dyadic half of $(0, t)$ by

$$\mathcal{K}_t = \frac{2}{t} \int_{t/2}^t P_h \, dh.$$

The uniform mixture $\bar{P}_H$ contains this component with weight $t/(2H)$ whenever $t \leq H$. The following elementary relation holds.

Lemma 3.1. *Let $0 < t_J < \dots < t_0 \leq H$, and suppose that the intervals $(t_j/2, t_j)$, $j \in \{0, \dots, J\}$, are pairwise disjoint. Then, for $f \in L^2(\pi)$,*

$$\mathcal{E}_{\bar{P}_H}(f, f) \geq \sum_{j=0}^J \frac{t_j}{2H} \mathcal{E}_{\mathcal{K}_{t_j}}(f, f).$$

Proof. By nonnegativity and Tonelli's theorem,

$$\mathcal{E}_{\bar{P}_H}(f, f) = \frac{1}{H} \int_0^H \mathcal{E}_{P_h}(f, f) \, dh \geq \sum_{j=0}^J \frac{1}{H} \int_{t_j/2}^{t_j} \mathcal{E}_{P_h}(f, f) \, dh.$$

The definition of $\mathcal{K}_{t_j}$ gives the claim. $\square$

We will analyze $\mathcal{K}_t$ at different scale parameters t , and aggregate results along an appropriate sequence of scale levels in a way that is more sophisticated than Lemma 3.1.

To be more specific, in Proposition B.1, we bound the fixed-step MALA rejection rate and use this bound to establish a close-coupling, or one-step overlap, result for $\mathcal{K}_t$ in Proposition 3.2. The curvature assumption on U yields the isoperimetric inequality in Proposition 3.3. We then combine Propositions 3.2 and 3.3 in Proposition 3.4 to bound the one-step flow for $\mathcal{K}_t$ out of a given set S . Importantly, for each set S , the flow out of S is controlled for one particular $\mathcal{K}_t$ where the scale parameter t depends on S . The general component-aggregation bound in Theorem 3.6 combines the one-step flows at various scales, yielding Theorem 2.1.

3.1 Close coupling

The next proposition, proved in Section B, shows that $\mathcal{K}_t(x, \cdot)$ and $\mathcal{K}_t(y, \cdot)$ are close in total variation whenever $\|x - y\| \leq \sqrt{t}/16$. For the larger admissible values of t , this conclusion is guaranteed only when x and y lie outside a small exceptional set.

Proposition 3.2. *There are universal constants $c_r, C_r > 0$ such that, for every real $p \geq 1$ and every t satisfying*

$$0 < t \leq \frac{c_r}{L\sqrt{p(d+p)}}, \quad (6)$$

there is a measurable set $G_{t,p} \subseteq \mathbb{R}^d$ satisfying

$$\pi(G_{t,p}^c) \leq \left[C_r L t \sqrt{p(d+p)} \right]^p, \quad (7)$$

and

$$x, y \in G_{t,p}, \quad \|x - y\| \leq \frac{\sqrt{t}}{16} \implies \text{TV}(\mathcal{K}_t(x, \cdot), \mathcal{K}_t(y, \cdot)) \leq \frac{3}{4}.$$

Moreover, if $t' \in (0, 1/(2Ld)]$, then

$$x, y \in \mathbb{R}^d, \quad \|x - y\| \leq \frac{\sqrt{t'}}{16} \implies \text{TV}(\mathcal{K}_{t'}(x, \cdot), \mathcal{K}_{t'}(y, \cdot)) \leq \frac{3}{4}.$$

3.2 Isoperimetric inequality

A strongly log-concave distribution satisfies the Gaussian enlargement inequality below; see [Bakry and Ledoux \[1996\]](#), Corollary 2.2 and equations (2.10)–(2.12)]. It remains valid under the first-order assumptions (3). To see this without assuming second differentiability, note that $U(x) - m \|x\|^2/2$ is convex. Caffarelli’s contraction theorem [[Caffarelli, 2000](#)], in the nonsmooth formulation of [Kim and Milman \[2012\]](#), Theorem 1.1, Lemma 3.3, and the conclusion of Section 3], therefore gives a 1-Lipschitz map T sending $N_d(0, m^{-1}I_d)$ to π . For each Borel set A and $r > 0$, the inclusion $(T^{-1}(A))^r \subseteq T^{-1}(A^r)$ transfers the Gaussian isoperimetric inequality to π . Consequently,

$$\pi(A^r) \geq \Phi(\Phi^{-1}(\pi(A)) + \sqrt{m}r), \quad (8)$$

where, for $r > 0$, $A^r = \{x : \text{dist}(x, A) < r\}$, and Φ is the standard Gaussian distribution function. The cases $\pi(A) = 0, 1$ are understood by continuity. The same bound holds for sets measurable in the completion of π , by choosing a Borel subset of the same measure. The following three-set form of the isoperimetric inequality is proved in [Section C](#).

Proposition 3.3. *Let A, B, C be a measurable partition of $\mathbb{R}^d$, let $r \geq 0$, and suppose $\text{dist}(A, B) \geq r$. If $0 < q = \min\{\pi(A), \pi(B)\} \leq 1/2$, then*

$$\pi(C) \geq \frac{q}{4} \min \left\{ 1, r \sqrt{m \log(1/q)} \right\}. \quad (9)$$

3.3 One-step flow

For a π -reversible Markov kernel K , define the one-step flow

$$\mathcal{J}_K(S, S^c) = \int_S K(x, S^c) \pi(dx).$$

Combining [Proposition 3.2](#) and [Proposition 3.3](#), we establish the following result concerning $\mathcal{K}_t$ in [Section D](#).

Proposition 3.4. *One can find universal constants $b_0 \in (0, 1/2]$ and $A_0 \in [1, \infty)$ such that the following holds. Let $p \geq p_\star := A_0[1 + \log(d+1) + \log \kappa]$, $\theta \in (0, 1]$, and*

$$t = \frac{\theta b_0}{L \sqrt{p(d+p)}}.$$

Then, for every measurable set $S \subseteq \mathbb{R}^d$ such that $e^{-p/2} \leq \pi(S) \leq 1/2$,

$$\mathcal{J}_{\mathcal{K}_t}(S, S^c) \geq 2^{-13} \pi(S) \sqrt{mt \log[1/\pi(S)]},$$

where $mt \log[1/\pi(S)] \leq 1$.

Moreover, for $t' \in (0, 1/(2Ld)]$ and every measurable set $S \subseteq \mathbb{R}^d$ such that $0 < \pi(S) \leq 1/2$,

$$\mathcal{J}_{\mathcal{K}_{t'}}(S, S^c) \geq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt' \log[1/\pi(S)]} \right\}.$$

Using [Proposition 3.4](#), for a set $S \subset \mathbb{R}^d$ satisfying $0 < \pi(S) < 1/2$, we may bound the average flow $\mathcal{J}_{\mathcal{K}_t}(S, S^c)/\pi(S)$ for appropriate values of t . The smaller $\pi(S)$ is, the smaller t for which the average flow can be estimated via [Proposition 3.4](#).

3.4 Aggregation of one-step flows

In [Section 4](#), we choose disjoint intervals of the form $[t_j/2, t_j]$ contained in $[0, H]$, where $(t_0, \dots, t_J)$ is a decreasing sequence of positive numbers. For a given measurable set S such that $0 < \pi(S) \leq 1/2$, the average flow $\mathcal{J}_{\mathcal{K}_{t_j}}(S, S^c)/\pi(S)$ may be large for some values of j but small for others. We therefore assign to each set S an appropriate index $j(S)$. Then, we establish a lower bound $\phi_{j(S)}$ on $\mathcal{J}_{\mathcal{K}_{t_{j(S)}}}(S, S^c)/\pi(S)$ using [Proposition 3.4](#). The purpose of this subsection is to provide the tools for combining these set-dependent bounds into a lower bound on $\text{Gap}_R(\bar{P}_H)$.

Cheeger's inequality [[Lawler and Sokal, 1988](#)] implies that the spectral gap of a reversible Markov kernel can be estimated by considering the kernel's average flow out of measurable sets. It may be tempting to use [Lemma 3.1](#) with $f = \mathbf{1}_S$, where $0 < \pi(S) \leq 1/2$, to obtain

$$\frac{\mathcal{J}_{\bar{P}_H}(S, S^c)}{\pi(S)} \geq \frac{1}{\pi(S)} \sum_{j=0}^J \frac{t_j}{2H} \mathcal{J}_{\mathcal{K}_{t_j}}(S, S^c) \geq \frac{t_{j(S)}}{2H} \frac{\mathcal{J}_{\mathcal{K}_{t_{j(S)}}}(S, S^c)}{\pi(S)} \geq \frac{\min_j t_j \phi_j}{2H}.$$

This gives a lower bound on the conductance of $\bar{P}_H$. It turns out that this direct estimate followed by the usual Cheeger inequality loses too much and does not yield the spectral lower bound of order $(\kappa\sqrt{d})^{-1}$ that we desire. We carry out the more careful approach below.

The following lemma carries out this aggregation for a generic mixture of reversible Markov kernels. It is possible to derive the lemma from the general Cheeger inequality of [Chen and Wang \[2000\]](#). We give a direct proof in [Section E](#).

Lemma 3.5. *Let $P, K_1, \dots, K_N$ be Markov kernels that are reversible with respect to a distribution ϖ . Suppose that there exist constants $\gamma_1, \dots, \gamma_N > 0$ such that*

$$\mathcal{E}_P(f, f) \geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(f, f) \quad \text{for every } f \in L^2(\varpi).$$

Suppose further that there exist constants $\beta_1, \dots, \beta_N \geq 0$ satisfying

$$\sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} > 0$$

such that, for every measurable set S satisfying $0 < \varpi(S) \leq 1/2$,

$$\sum_{j=1}^N \beta_j \mathcal{J}_{K_j}(S, S^c) \geq \varpi(S),$$

where

$$\mathcal{J}_{K_j}(S, S^c) = \int_S K_j(x, S^c) \varpi(dx).$$

Then

$$\text{Gap}_R(P) \geq \left(2 \sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} \right)^{-1}.$$

The above lemma gives the following convenient hard-assignment formulation, which we use to combine the flow bounds in [Proposition 3.4](#). Roughly speaking, [Theorem 3.6](#) allows the kernel that supplies the useful one-step flow bound to depend on the set under consideration, while keeping track of how often each kernel is selected in the mixture.

Theorem 3.6. *Let $P, K_1, \dots, K_N$ be Markov kernels that are reversible with respect to a distribution ϖ . Suppose*

$$\mathcal{E}_P(f, f) \geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(f, f), \quad \gamma_j > 0,$$

for every $f \in L^2(\varpi)$.

Suppose further that there exist constants $\phi_1, \dots, \phi_N > 0$ such that every measurable set S with $0 < \varpi(S) \leq 1/2$ can be assigned an index $j(S) \in \{1, \dots, N\}$ satisfying

$$\mathcal{J}_{K_{j(S)}}(S, S^c) \geq \phi_{j(S)} \varpi(S).$$

Then

$$\text{Gap}_R(P) \geq \left(2 \sum_{j=1}^N \frac{1}{\gamma_j \phi_j^2} \right)^{-1}.$$

Proof. Apply Lemma 3.5 with

$$\beta_j = \frac{1}{\phi_j}, \quad j = 1, \dots, N.$$

For every measurable S with $0 < \varpi(S) \leq 1/2$,

$$\begin{aligned} \sum_{j=1}^N \beta_j \mathcal{J}_{K_j}(S, S^c) &\geq \frac{\mathcal{J}_{K_{j(S)}}(S, S^c)}{\phi_{j(S)}} \\ &\geq \varpi(S). \end{aligned}$$

The conclusion follows directly from Lemma 3.5. □

4 Proof of the Main Theorem

Proof of Theorem 2.1. Let $b_0 \leq 1/2$ and $A_0 \geq 1$ be given in Proposition 3.4, and recall that $p_\star = A_0[1 + \log(d+1) + \log \kappa]$. We divide the proof into three parts.

Part 1: The geometric moment ladder.

Suppose for now that $p_\star < d$. Let

$$p_j = 4^j p_\star, \quad j = 0, 1, \dots, J, \tag{10}$$

where J is the smallest integer for which $p_J \geq d$. Thus

$$d \leq p_J < 4d. \tag{11}$$

Define the nominal endpoints

$$\tau_j = \frac{b_0}{L \sqrt{p_j(d + p_j)}},$$

and put

$$\theta = \min \left\{ 1, \frac{H}{\tau_0} \right\}, \quad t_j = \theta \tau_j. \tag{12}$$

Then $t_0 \leq H$, and

$$\frac{t_{j+1}}{t_j} = \frac{1}{2} \sqrt{\frac{d + p_j}{d + 4p_j}} < \frac{1}{2}.$$

Consequently, the Borel sets $[t_j/2, t_j]$ are pairwise disjoint, have Lebesgue measure $t_j/2$, and lie within $[0, H]$. Applying [Lemma 3.1](#) to this finite family gives

$$\mathcal{E}_{\overline{P}_H}(f, f) \geq \sum_{j=0}^J \frac{t_j}{2H} \mathcal{E}_{\mathcal{K}_{t_j}}(f, f). \quad (13)$$

This prompts us to apply [Theorem 3.6](#).

Let S be measurable with $0 < \pi(S) \leq 1/2$. We assign S to a component $\mathcal{K}_{t_j}$. That is, we identify the index $j(S)$ in [Theorem 3.6](#). Since $A_0 \geq 1$, $d \geq 1$, and $\kappa \geq 1$,

$$\frac{p_0}{2} = \frac{p_\star}{2} \geq \frac{1 + \log 2}{2} > \log 2.$$

The intervals $[\log 2, p_0/2]$, $(p_{j-1}/2, p_j/2]$ for $1 \leq j \leq J$, and $(p_J/2, \infty)$ are disjoint and cover $[\log 2, \infty)$. Thus the following assignment is exhaustive and unambiguous.

- If $\log 2 \leq \log[1/\pi(S)] \leq p_0/2$, assign S to $\mathcal{K}_{t_0}$. By the first assertion of [Proposition 3.4](#),

$$\mathcal{J}_{\mathcal{K}_{t_0}}(S, S^c) \geq \phi_0 \pi(S), \quad \phi_0 = 2^{-13} \sqrt{mt_0 \log 2}.$$

- For $1 \leq j \leq J$, if $p_{j-1}/2 < \log[1/\pi(S)] \leq p_j/2$, assign S to $\mathcal{K}_{t_j}$. By the first assertion of [Proposition 3.4](#),

$$\mathcal{J}_{\mathcal{K}_{t_j}}(S, S^c) \geq \phi_j \pi(S), \quad \phi_j = 2^{-13} \sqrt{\frac{mt_j p_{j-1}}{2}} = 2^{-13} \sqrt{\frac{mt_j p_j}{8}}.$$

- If $\log[1/\pi(S)] > p_J/2$, assign S to the last component $\mathcal{K}_{t_J}$. By [\(11\)](#) and the assumption $b_0 \leq 1/2$,

$$t_J \leq \frac{b_0}{L\sqrt{p_J(d+p_J)}} \leq \frac{1}{2Ld}.$$

By the second assertion of [Proposition 3.4](#),

$$\mathcal{J}_{\mathcal{K}_{t_J}}(S, S^c) \geq \phi'_J \pi(S), \quad \phi'_J = 2^{-13} \min \left\{ 1, \sqrt{\frac{mt_J p_J}{2}} \right\}.$$

Moreover,

$$\frac{mt_J p_J}{2} \leq \frac{b_0}{2\kappa} \sqrt{\frac{p_J}{d+p_J}} \leq \frac{b_0}{2} < 1.$$

Weakening the flow bound by a constant factor, we obtain

$$\mathcal{J}_{\mathcal{K}_{t_J}}(S, S^c) \geq \phi_J \pi(S), \quad \phi_J = 2^{-13} \sqrt{\frac{mt_J p_J}{8}}.$$

We may therefore apply [Theorem 3.6](#) to [\(13\)](#), with

$$\gamma_j = \frac{t_j}{2H}, \quad \phi_0^2 = 2^{-26} mt_0 \log 2, \quad \phi_j^2 = 2^{-29} mt_j p_j \quad (j \geq 1).$$

The summation estimate in [Lemma F.1](#) yields

$$\sum_{j=0}^J \frac{1}{\gamma_j \phi_j^2} \leq C \frac{HL^2 p_\star (d + p_\star)}{m\theta^2 b_0^2},$$

where $C \in (0, \infty)$ is universal. Hence,

$$\text{Gap}_R(\bar{P}_H) \geq \frac{m\theta^2 b_0^2}{2CHL^2 p_*(d+p_*)} = \frac{m}{2CH} \min \left\{ H, \frac{b_0}{L\sqrt{p_*(d+p_*)}} \right\}^2. \quad (14)$$

Part 2: A conservative bound

The bound (14) is almost what we need, but it has been established only for $p_* < d$. To obtain the exact bound in [Theorem 2.1](#), we derive a lower bound on $\text{Gap}_R(\bar{P}_H)$ using only conservative step sizes.

Define the safe step size

$$t_s = \min \left\{ H, \frac{b_0}{Ld} \right\}.$$

Because $b_0 \leq 1/2$, $t_s \leq 1/(2Ld)$. For every measurable S with $0 < \pi(S) \leq 1/2$, apply the second assertion of [Proposition 3.4](#) to obtain

$$\mathcal{J}_{\mathcal{K}_{t_s}}(S, S^c) \geq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt_s \log[1/\pi(S)]} \right\} \geq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt_s \log 2} \right\}. \quad (15)$$

Since

$$mt_s \log 2 \leq \frac{b_0 \log 2}{\kappa d} < 1,$$

the bound (15) becomes

$$\mathcal{J}_{\mathcal{K}_{t_s}}(S, S^c) \geq 2^{-13} \pi(S) \sqrt{mt_s \log 2}.$$

By [Lemma 3.1](#),

$$\mathcal{E}_{\bar{P}_H}(f, f) \geq \frac{t_s}{2H} \mathcal{E}_{\mathcal{K}_{t_s}}(f, f).$$

Apply [Theorem 3.6](#) with $N = 1$, $K_1 = \mathcal{K}_{t_s}$, $\gamma_1 = t_s/(2H)$, and $\phi_1 = 2^{-13} \sqrt{mt_s \log 2}$ to obtain

$$\text{Gap}_R(\bar{P}_H) \geq 2^{-28} (\log 2) \frac{mt_s^2}{H} = 2^{-28} (\log 2) \frac{m}{H} \min \left\{ H, \frac{b_0}{Ld} \right\}^2. \quad (16)$$

Part 3: Completion of the proof

If $p_* \geq d$, then

$$\frac{1}{\sqrt{p_*(d+p_*)}} \leq \frac{1}{d},$$

so the safe bound (16) already gives (4). If $p_* < d$, combine (14) and (16). For positive a, b, H ,

$$\max\{\min(H, a)^2, \min(H, b)^2\} = \min\{H, \max(a, b)\}^2.$$

Then (4) follows.

It remains to verify the assertion about lazification. Since the identity kernel has zero Dirichlet form,

$$\mathcal{E}_{(I+\bar{P}_H)/2}(f, f) = \frac{1}{2} \mathcal{E}_{\bar{P}_H}(f, f),$$

so

$$\text{Gap}_R((I + \bar{P}_H)/2) = \frac{1}{2} \text{Gap}_R(\bar{P}_H).$$

□

5 Lean verification

The accompanying development in Lean 4 [de Moura and Ullrich, 2021] provides proof scripts for the principal results of this article. Its first-order input consists of a continuously differentiable potential, the strong-convexity inequality in (3), and a Lipschitz bound on the actual gradient of that potential.

The supplied scripts connect this input to both the non-lazy and half-lazy conclusions of Theorem 2.1, with the same universal constants in the two conclusions, and to the first two bounds in Corollary 2.2. The smooth hard-target construction and the fixed-step minimax statement in Propositions A.1 and 2.3 are retained. The randomized-step proof uses the mixture comparison in Lemma 3.1, the overlap and separation bounds in Propositions 3.2 and 3.3, both flow bounds in Proposition 3.4, and the aggregation results in Lemma 3.5 and Theorem 3.6.

The stationary-rejection and overlap conclusions (Propositions B.1 and 3.2) are obtained through an alternative discrete-time proof, using finite-dimensional Gaussian likelihood estimates, an Euler/random-walk-Metropolis comparison, and weak-limit stability. The continuous-time proofs of Lemmas B.2 to B.5 are not transcribed into Lean. In particular, the additional nonconvex scope noted at the beginning of Appendix B is not part of the formalization: the Lean stationary-rejection theorem retains the standing strong-convexity assumption.

The isoperimetric bound is derived internally from the Gaussian Ornstein–Uhlenbeck argument and a finite-Euler weak limit, rather than assuming the contraction theorem cited in Section 3.2. The development also contains reusable results on first-order Taylor bounds, equivalence of the L^2 Poincaré and Rayleigh formulations, reversibility and Dirichlet forms of kernel mixtures, lazification, flow aggregation, spectral-gap upper bounds, and weak-limit stability.

The sole direct external Lean library is mathlib [The mathlib Community, 2020]; its standard dependencies and Lean’s logical foundations are retained. The revised source contains no proof placeholders or paper-specific axiom declarations.

The complete development was kernel-checked with Lean 4.33.0 and mathlib 4.33.0; the full build and the axiom audit of 266 principal declarations succeeded, with no proof placeholders or project-specific axioms.

6 Numerical experiments

We compare fixed-step MALA and uniformly randomized-step MALA on a perturbed-Gaussian target. The simulation package can be found in a GitHub repository [Qin, 2026].

Let h_0 be a positive number and consider a potential function

$$U_{d,h_0}(x) = \frac{m}{2}x_1^2 + \sum_{i=2}^d \left[\frac{L+m}{4}x_i^2 - \frac{(L-m)h_0}{2} \cos\left(\frac{x_i}{\sqrt{h_0}}\right) \right]. \quad (17)$$

In Section A, perturbed Gaussian distributions with densities proportional to $e^{-U_{d,h}}$ are investigated from a theoretical perspective. As shown in Proposition A.1, $mI_d \preceq \nabla^2 U_{d,h_0}(x) \preceq LI_d$ for every x . Throughout, $m = 0.1$ and $L = 1$, so $\kappa = 10$. This type of construction appears in [Chewi et al., 2021, Lee et al., 2021, Wu et al., 2022], and it serves as a hard target for fixed-step MALA.

For an endpoint $H > 0$, we compare a fixed-step MALA P_h with $h = H$ and the randomized-step $\bar{P}_H$. We report the mean acceptance probability $\mathbb{E}[\alpha_h(X, Y)]$ and the expected squared jumping distance per coordinate,

$$\text{ESJD} = \frac{1}{d} \mathbb{E} \left[\alpha_h(X, Y) \|Y - X\|^2 \right],$$

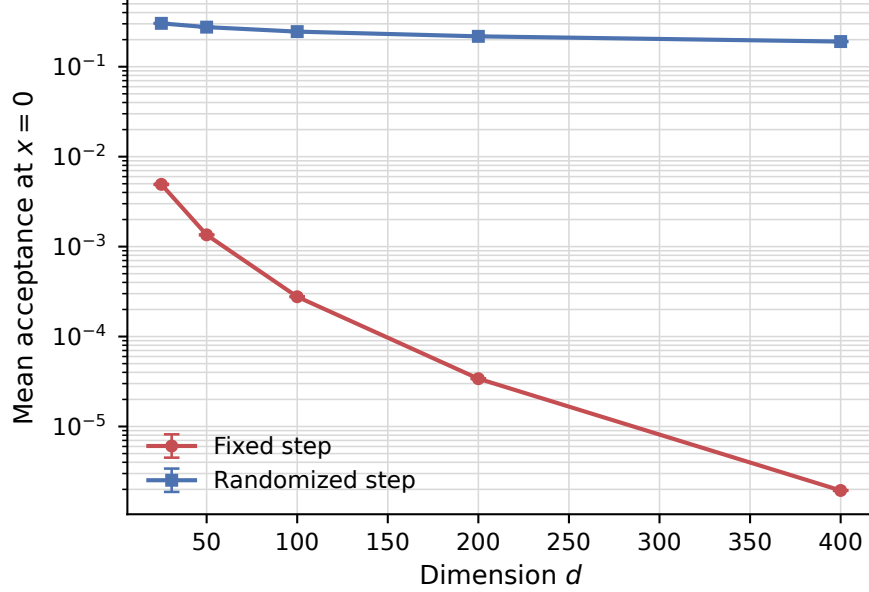


Figure 1: Estimated mean acceptance probability at the origin for the target (17). Fixed-step MALA uses $h = h_0$, while randomized-step MALA uses $h \sim \text{Unif}(0, h_0)$. The vertical axis is logarithmic, and vertical bars show ± 2 estimated standard errors; most are narrower than the markers.

where X is the current state, and Y is the proposed state. For fixed-step MALA, $Y \mid X, h \sim Q_h(X, \cdot)$ where $h = H$ is fixed. For randomized-step MALA, $Y \mid X, h \sim Q_h(X, \cdot)$ and $h \sim \text{Unif}(0, H)$. We will investigate the setting $X = 0$ as well as $X \sim \pi$. To estimate the mean acceptance rate and the ESJD, we use classical Monte Carlo. That is, independent copies of (X, Y, h) are generated, and expectations under the empirical distribution are used as estimates.

6.1 Acceptance at the origin

For each $d \in \{25, 50, 100, 200, 400\}$, we set

$$h_0 = \frac{4}{L\sqrt{d}}$$

and compare fixed-step MALA with $h = h_0$ against randomized-step MALA with $H = h_0$ in terms of the mean acceptance rate at the origin. As shown in Section A, the origin is where fixed-step MALA struggles to leave. Each estimate uses 80,000 independent proposals.

As Figure 1 shows, the fixed-step acceptance probability decreases from approximately 4.92×10^{-3} at $d = 25$ to 1.94×10^{-6} at $d = 400$. The randomized-step acceptance probability decreases much more slowly, from 0.304 to 0.190. Because rejection leaves a chain initialized at the origin unchanged, the time to its first accepted move is geometric. At $d = 400$, the reciprocal acceptance estimates imply mean waiting times of approximately 5.14×10^5 iterations for fixed-step MALA and 5.25 iterations for randomized-step MALA. The experiment therefore directly displays the sticky behavior behind the fixed-step obstruction: proposals at the oscillation scale h_0 have rapidly deteriorating acceptance, whereas uniform randomization retains a substantial probability of selecting a smaller step.

To continue the origin experiment beyond one step, we next examine what happens after fixed-step MALA eventually escapes the hard point. We run one chain of each type for 40,000 iterations

on the $d = 200$ target, starting both chains at the origin and taking $H = h_0$. We record the first coordinate at every iteration.

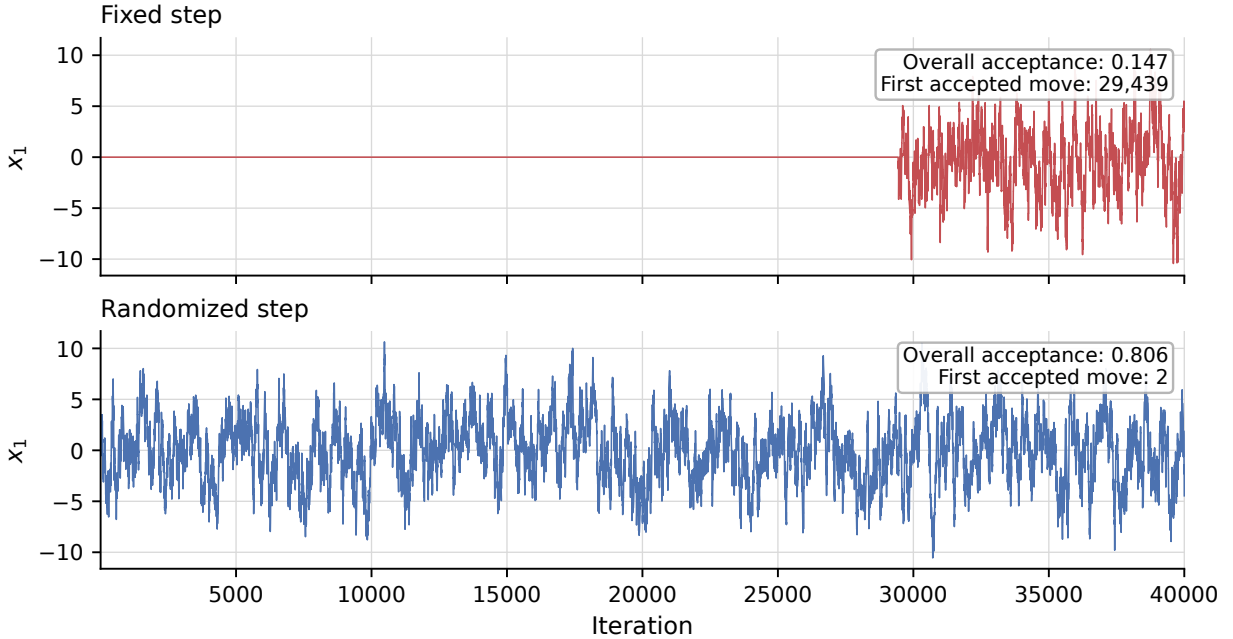


Figure 2: First-coordinate traces continuing the origin experiment for the $d = 200$ hard target, with both chains initialized at the origin and $H = h_0 = 4/\sqrt{200}$. The two panels use common horizontal and vertical scales.

In this realization, the fixed-step chain rejects its first 29,438 proposals and first moves at iteration 29,439. It then makes 5,882 accepted moves by iteration 40,000, so its overall acceptance rate is 0.147. Over the 10,562-iteration interval from its first accepted move through the end of the run, its empirical acceptance rate is 0.557. The trace therefore shows that, once the fixed chain escapes the sticky point, it begins moving on an ordinary scale; the low overall acceptance rate is dominated by the long initial wait. The randomized chain first moves at iteration 2, makes 32,253 accepted moves, and has overall acceptance rate 0.806.

6.2 Stationary acceptance and jumping distance

The preceding experiment deliberately starts at an atypical point. We next compare average one-step behavior under stationarity. We fix $d = 200$ and $h_0 = 4/\sqrt{200}$, and vary H/h_0 from 0.05 to 100, using a fine grid near the efficient range and a logarithmically spaced grid on the over-tuning side.

Exact stationary samples are available because (17) is a product target. The first coordinate is $N(0, m^{-1})$. For each remaining coordinate, we propose $\tilde{X} \sim N(0, 2/(L + m))$ and accept it with probability

$$\exp \left\{ \frac{(L - m)h_0}{2} \left[\cos \left(\frac{\tilde{X}}{\sqrt{h_0}} \right) - 1 \right] \right\}.$$

This is an exact rejection sampler. At every endpoint H and for each method, we use 48,000 independent stationary one-step experiments.

Figure 3 shows the expected efficiency–robustness tradeoff. At $H = h_0$, fixed-step MALA has acceptance 0.559 and ESJD 0.340, while randomized-step MALA has acceptance 0.804 and ESJD

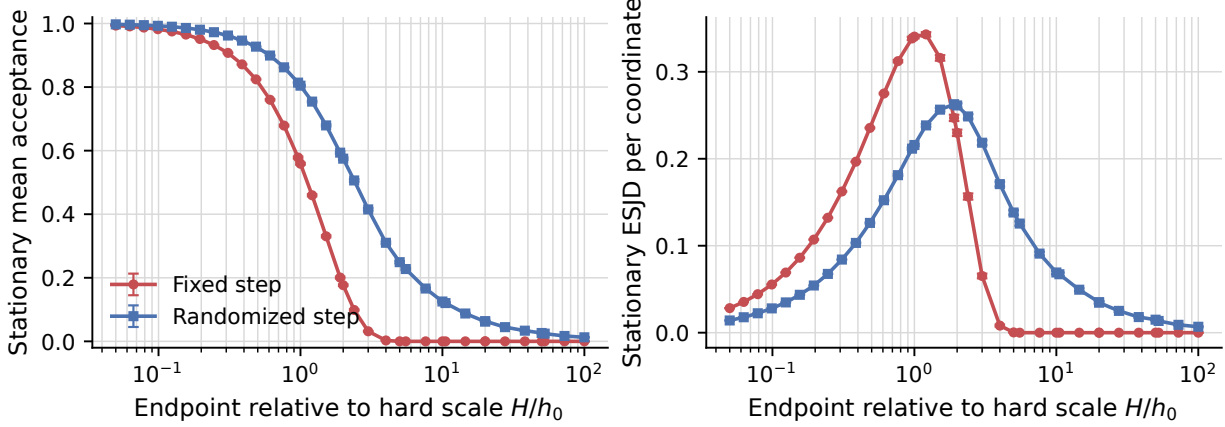


Figure 3: Stationary one-step performance for the $d = 200$ target with $h_0 = 4/\sqrt{200}$. Fixed-step MALA uses $h = H$, while randomized-step MALA uses $h \sim \text{Unif}(0, H)$. Vertical bars show ± 2 estimated standard errors.

0.216. The largest observed fixed-step ESJD is 0.343, at $H/h_0 \approx 1.21$, whereas the largest observed randomized-step ESJD is 0.263, at $H/h_0 \approx 1.90$. Thus a well-tuned fixed step has the larger peak ESJD.

Randomization becomes advantageous when the endpoint is too large. At $H = 10h_0$, fixed-step acceptance is extremely close to 0, whereas randomized-step MALA retains acceptance 0.125 and ESJD 0.0692. Even at $H = 100h_0$, the randomized method has acceptance 0.0127 and ESJD 0.00681, while the corresponding fixed-step estimates are below floating-point resolution. Randomization does not prevent performance from eventually decreasing as H grows, but it changes the abrupt fixed-step collapse into a much more gradual decline.

The two experiments also expose a distinction relevant to the global spectral gap. When $d = 200$ and $H = h_0$, fixed-step acceptance is about 0.559 under stationarity but only 3.40×10^{-5} at the origin. Thus stationary-average diagnostics largely miss the small sticky region, which leads to the exceedingly small spectral gap of fixed-step MALA, as shown in [Section A](#). Uniform randomization improves acceptance both in that region and under over-tuning, although it does not dominate a well-tuned fixed step in stationary ESJD. These one-step experiments illustrate tuning robustness; they do not estimate the spectral gap itself.

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Appendices

A Proof of Proposition 2.3

We follow an existing line of research on hard targets for fixed-step MALA. Perturbed-Gaussian constructions appear in [Chewi et al. \[2021, Theorems 8, 9, and 36\]](#). The product structure combining a Gaussian coordinate with cosine-perturbed Gaussian coordinates appears in [Lee et al. \[2021,](#)

Supplement, equation (20)] and [Wu et al. \[2022, Equation \(28\) and Lemma 8\]](#). Below, we use a modified version of the latter construction to obtain the explicit bound in [Proposition 2.3](#).

Proposition A.1. *There are universal constants $c, C > 0$ such that the following holds. Let $d \geq 2$, $0 < m < L$, and $h > 0$. Define*

$$U_{d,h}(x) = \frac{m}{2}x_1^2 + \sum_{i=2}^d \left[\frac{L+m}{4}x_i^2 - \frac{(L-m)h}{2} \cos\left(\frac{x_i}{\sqrt{h}}\right) \right]. \quad (18)$$

Then $U_{d,h}$ is infinitely differentiable and

$$mI_d \preceq \nabla^2 U_{d,h}(x) \preceq LI_d, \quad x \in \mathbb{R}^d. \quad (19)$$

If P_h is the MALA kernel with step size h and target density proportional to $e^{-U_{d,h}}$, then

$$\text{Gap}_R(P_h) \leq C \min \left\{ mh + \frac{(mh)^2}{2}, \exp[-c(d-1) \min\{(L-m)h, 1\}] \right\}. \quad (20)$$

Writing $\kappa = L/m$, this further implies, after possibly changing the universal constants c and C , that

$$\text{Gap}_R(P_h) \leq C \min \left\{ \kappa^{-1}Lh, \exp[-cd \min\{(1-\kappa^{-1})Lh, 1\}] \right\}. \quad (21)$$

Proof. For $i \geq 2$, the i th diagonal entry of the Hessian of $U_{d,h}$ is

$$\frac{L+m}{2} + \frac{L-m}{2} \cos\left(\frac{x_i}{\sqrt{h}}\right),$$

which lies in $[m, L]$. The first diagonal entry equals m , and all off-diagonal entries vanish. This proves (19).

We first establish the local upper bound. The target is a product measure, and its first marginal is $N(0, m^{-1})$. Use $f(x) = x_1$ in the Poincaré quotient. Let X have the target distribution, let $Z \sim N_d(0, I_d)$ be independent of X , and write

$$\tilde{Y} = X - h\nabla U_{d,h}(X) + \sqrt{2h}Z$$

for the MALA proposal. The actual transition either moves to $\tilde{Y}$ or remains at X . Consequently,

$$\begin{aligned} \mathcal{E}_{P_h}(f, f) &\leq \frac{1}{2} \mathbb{E} \left[(-hmX_1 + \sqrt{2h}Z_1)^2 \right] \\ &= \frac{1}{2} (h^2m + 2h), \end{aligned}$$

where we used $\mathbb{E}(X_1^2) = m^{-1}$ and the independence of X_1 and Z_1 . Since $\text{Var}_\pi(f) = m^{-1}$, it follows that

$$\text{Gap}_R(P_h) \leq mh + \frac{(mh)^2}{2}. \quad (22)$$

We next obtain the exponential upper bound by controlling the mean acceptance probability at the origin. Write

$$a = \frac{L+m}{2}, \quad b = \frac{L-m}{2}.$$

At the origin, $\nabla U_{d,h}(0) = 0$, so a proposal has the form

$$Y = \sqrt{2h}Z.$$

For a general differentiable potential U , the log Hastings ratio for a proposal from x to y has the following standard representation; see also [Eberle \[2014, Proposition 1.3, equation \(1.33\)\]](#).

$$\begin{aligned} R_h(x, y) &:= \log \frac{e^{-U(y)} q_h(y, x)}{e^{-U(x)} q_h(x, y)} \\ &= U(x) - U(y) + \frac{1}{2} \langle y - x, \nabla U(x) + \nabla U(y) \rangle \\ &\quad + \frac{h}{4} \left[\|\nabla U(x)\|^2 - \|\nabla U(y)\|^2 \right]. \end{aligned} \tag{23}$$

For $i = 2, \dots, d$, put

$$V_i = \frac{Y_i}{\sqrt{h}} = \sqrt{2} Z_i.$$

Substituting (18) into (23) gives

$$\begin{aligned} R_h(0, Y) &= -\frac{m^2 h^2}{2} Z_1^2 + b h \sum_{i=2}^d A(V_i) \\ &\quad - \frac{h^2}{4} \sum_{i=2}^d [a V_i + b \sin(V_i)]^2, \end{aligned} \tag{24}$$

where

$$A(v) = \cos v - 1 + \frac{v \sin v}{2}.$$

Both the first and the last terms in (24) are nonpositive.

Since $V_i \sim N(0, 2)$,

$$\mathbb{E}(\cos V_i) = e^{-1}, \quad \mathbb{E}(V_i \sin V_i) = 2e^{-1}.$$

The first identity follows from the characteristic function of V_i , and the second follows from Stein's identity. Hence

$$\mathbb{E}[A(V_i)] = \frac{2}{e} - 1 = -a_0, \quad a_0 = 1 - \frac{2}{e} > 0.$$

Moreover,

$$|A(v)| \leq 2 + \frac{|v|}{2}.$$

Thus $A(V_i) - \mathbb{E}[A(V_i)]$ has a universally bounded sub-Gaussian norm; see, for example, [\[Vershynin, 2018, Section 2.5.2\]](#). The concentration inequality for sums of independent sub-Gaussian random variables [\[Vershynin, 2018, Theorem 2.6.3\]](#) therefore yields a universal constant $c_0 > 0$ such that

$$\mathbb{P} \left(\sum_{i=2}^d A(V_i) > -\frac{a_0(d-1)}{2} \right) \leq e^{-c_0(d-1)}. \tag{25}$$

Outside the event in (25),

$$R_h(0, Y) \leq -\frac{a_0}{2} b h (d-1) = -\frac{a_0}{4} (d-1) (L-m) h.$$

Let

$$a_h(x) = \int \alpha_h(x, y) Q_h(x, dy)$$

denote the pointwise acceptance probability. It follows that

$$\begin{aligned} a_h(0) &= \mathbb{E} \left[1 \wedge e^{R_h(0,Y)} \right] \\ &\leq \exp \left[-\frac{a_0}{4}(d-1)(L-m)h \right] + e^{-c_0(d-1)} \\ &\leq 2 \exp [-c_1(d-1) \min\{(L-m)h, 1\}] \end{aligned}$$

for a universal constant $c_1 > 0$.

The function $x \mapsto a_h(x)$ is continuous. Indeed, writing the proposal as

$$x - h \nabla U_{d,h}(x) + \sqrt{2h} Z, \quad Z \sim N_d(0, I_d),$$

the resulting acceptance probability is continuous in x for every fixed Z and is bounded by one. Continuity therefore follows from dominated convergence.

Consequently, there is an open ball S centered at the origin such that

$$0 < \pi(S) \leq \frac{1}{2}, \quad \sup_{x \in S} a_h(x) \leq 4 \exp [-c_1(d-1) \min\{(L-m)h, 1\}].$$

A transition starting in S can leave S only if its proposal is accepted. Hence

$$\mathcal{J}_{P_h}(S, S^c) \leq 4 \exp [-c_1(d-1) \min\{(L-m)h, 1\}] \pi(S).$$

Testing the Poincaré quotient with $\mathbf{1}_S$ gives

$$\begin{aligned} \text{Gap}_R(P_h) &\leq \frac{\mathcal{E}_{P_h}(\mathbf{1}_S, \mathbf{1}_S)}{\text{Var}_\pi(\mathbf{1}_S)} \\ &= \frac{\mathcal{J}_{P_h}(S, S^c)}{\pi(S)[1 - \pi(S)]} \\ &\leq 8 \exp [-c_1(d-1) \min\{(L-m)h, 1\}]. \end{aligned} \tag{26}$$

Combining (22) and (26) proves (20).

Finally, we prove (21). We have

$$L - m = (1 - \kappa^{-1})L, \quad d - 1 \geq \frac{d}{2}.$$

Therefore

$$(d-1) \min\{(L-m)h, 1\} \geq \frac{d}{2} \min\{(1 - \kappa^{-1})Lh, 1\}.$$

If $mh \leq 1$, then

$$mh + \frac{(mh)^2}{2} \leq \frac{3}{2}mh = \frac{3}{2}\kappa^{-1}Lh.$$

If $mh > 1$, then

$$\exp \left[-cd \min\{(1 - \kappa^{-1})Lh, 1\} \right] \leq 1 < mh = \kappa^{-1}Lh,$$

so the exponential bound is the smaller of the two terms in (21). After adjusting the universal constants, this proves (21). $\square$

We can now prove [Proposition 2.3](#).

Proof. By [Proposition A.1](#), there are universal constants $c_0, C_0 > 0$ such that, for every $h > 0$,

$$\inf_{\substack{U \in C^\infty(\mathbb{R}^d) \\ mI_d \preceq \nabla^2 U(x) \preceq LI_d, x \in \mathbb{R}^d}} \text{Gap}_R(P_{U,h}) \leq F(Lh), \quad (27)$$

where

$$F(t) = C_0 \min \left\{ \kappa^{-1}t, \exp \left[-c_0 d \min \{ (1 - \kappa^{-1})t, 1 \} \right] \right\}.$$

Set

$$t_\star = \frac{2 \log(\kappa d)}{c_0(1 - \kappa^{-1})d}.$$

If $0 < t \leq t_\star$, then

$$F(t) \leq C_0 \kappa^{-1} t_\star = \frac{2C_0 \log(\kappa d)}{c_0(1 - \kappa^{-1})\kappa d}.$$

If $t > t_\star$, then the exponential term defining F is nonincreasing in t , and hence

$$\begin{aligned} F(t) &\leq C_0 \exp \left[-c_0 d \min \{ (1 - \kappa^{-1})t_\star, 1 \} \right] \\ &= C_0 \max \left\{ \frac{1}{(\kappa d)^2}, e^{-c_0 d} \right\}. \end{aligned}$$

Consequently,

$$\begin{aligned} \sup_{t>0} F(t) &\leq C_0 \max \left\{ \frac{2 \log(\kappa d)}{c_0(1 - \kappa^{-1})\kappa d}, \frac{1}{(\kappa d)^2}, e^{-c_0 d} \right\} \\ &\leq C_0 \max \left\{ \frac{2 \log(\kappa d)}{c_0(1 - \kappa_0^{-1})\kappa d}, \frac{1}{(\kappa d)^2}, e^{-c_0 d} \right\}. \end{aligned} \quad (28)$$

Because $d \geq 2$ and $\kappa > 1$,

$$\frac{1}{(\kappa d)^2} \leq C \frac{\log(\kappa d)}{\kappa d}$$

for a universal constant C . Absorbing the factors depending only on κ_0 into C , and changing the universal exponential constant if necessary, (27) and (28) prove (5). $\square$

B Proof of Proposition 3.2

This appendix proves [Proposition 3.2](#). The convexity assumption on U is not needed for [Proposition B.1](#) and [Lemmas B.2 to B.5](#). For these results it suffices that $U \in C^1(\mathbb{R}^d)$, that ∇U be globally L -Lipschitz for some $L > 0$, and that $0 < \int_{\mathbb{R}^d} e^{-U(x)} dx < \infty$, with π the normalized measure and $X_0 \sim \pi$. Under these weaker assumptions the Langevin diffusion is still nonexplosive and reversible with respect to π ; see [Durmus and Moulines \[2017, Section 2, assumption L1 and the subsequent discussion\]](#). The proofs of the stated results below use only these weaker assumptions. The globally safe assertion of [Proposition 3.2](#), proved at the end of this appendix, continues to use convexity.

The main input is the following stationary estimate for the pointwise rejection function

$$r_h(x) = 1 - \int \alpha_h(x, y) Q_h(x, dy),$$

where

$$\alpha_h(x, y) = 1 \wedge \frac{e^{-U(y)} q_h(y, x)}{e^{-U(x)} q_h(x, y)}, \quad q_h(x, y) = (4\pi h)^{-d/2} \exp \left\{ -\frac{\|y - x + h \nabla U(x)\|^2}{4h} \right\}.$$

Proposition B.1. *There are universal constants $c, C > 0$ such that, for every real $p \geq 1$,*

$$h \leq \frac{c}{L\sqrt{p(d+p)}} \implies \|r_h\|_{L^p(\pi)} \leq CLh\sqrt{p(d+p)}. \quad (29)$$

The proof of [Proposition B.1](#) compares the exact stationary Langevin path with the path whose drift is frozen at its initial position. The endpoint of the latter path is exactly the MALA proposal. Similar diffusion-comparison ideas appear in [Bou-Rabee and Vanden-Eijnden \[2010\]](#), [Bou-Rabee and Hairer \[2013\]](#), [Chewi et al. \[2021\]](#).

The idea behind the proof is roughly as follows. Let $X_0 \sim \pi$, and suppose that, on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$, X_t solves the Langevin equation

$$dX_t = -\nabla U(X_t) dt + \sqrt{2} dB_t, \quad (30)$$

where B_t is a Brownian motion. Standard theory of Langevin diffusion implies $X_h \sim \pi$. Denote by S_h the conditional distribution of X_h given X_0 . We compare $S_h(x, dy)$ to the proposal $Q_h(x, dy)$, which gives the Gaussian law $N_d(x - h\nabla U(x), 2hI_d)$. Define $\nu_h(dx, dy) = \pi(dx)S_h(x, dy)$, $\mu_h(dx, dy) = \pi(dx)Q_h(x, dy)$. Assume that the Radon-Nikodym derivative $F_h = d\mu_h/d\nu_h$ can be found. Since the Langevin diffusion is π -reversible, it holds that ν_h is symmetric, and

$$\frac{d\mu_h^\top}{d\mu_h}(x, y) = \frac{F_h(y, x)}{F_h(x, y)},$$

where $\mu_h^\top(dx, dy) = \mu_h(dy, dx)$. Moreover, it can be argued that, for π -a.e. x , $F_h(x, \cdot) = dQ_h(x, \cdot)/dS_h(x, \cdot)$. Then

$$\begin{aligned} r_h(x) &= \int S_h(x, dy) \left[1 - F_h(x, y) \min \left\{ 1, \frac{F_h(y, x)}{F_h(x, y)} \right\} \right] \\ &\leq \int (|F_h(x, y) - 1| + |F_h(y, x) - 1|) S_h(x, dy). \end{aligned}$$

Using Jensen's inequality and the fact that ν_h is symmetric, one can now show that $\|r_h\|_{L^p(\pi)}$ can be controlled by $2\|F_h - 1\|_{L^p(\nu_h)}$.

To find F_h , let

$$dW_t = dB_t - \theta_t dt, \quad \theta_t = \frac{\nabla U(X_t) - \nabla U(X_0)}{\sqrt{2}}$$

Then X_t has the frozen drift representation $dX_t = -\nabla U(X_0)dt + \sqrt{2}dW_t$. Suppose that one can find an alternative probability measure $\mathbb{P}_h^{\text{fr}}$ defined by a Radon-Nikodym derivative $D_h = d\mathbb{P}_h^{\text{fr}}/d\mathbb{P}|_{\mathcal{F}_h}$ such that, under $\mathbb{P}_h^{\text{fr}}$, W_t is a Brownian motion with respect to $(\mathcal{F}_t)_{0 \leq t \leq h}$. Then, under this new probability measure, (X_0, X_h) is distributed as μ_h . In particular, with expectations defined in terms of $\mathbb{P}$, $F_h(X_0, X_h) = \mathbb{E}[D_h | \sigma(X_0, X_h)]$, so conditional Jensen's inequality would imply that $\|F_h - 1\|_{L^p(\nu_h)} \leq \|D_h - 1\|_{L^p(\mathbb{P})}$. As stated in [Lemma B.4](#), one can construct the desired measure $\mathbb{P}_h^{\text{fr}}$ by letting D_h be defined through the stochastic exponential

$$D_t = \exp \left(\int_0^t \langle \theta_s, dB_s \rangle - \frac{1}{2} V_t \right), \quad V_t = \int_0^t \|\theta_s\|^2 ds,$$

provided that $(D_t)_{0 \leq t \leq h}$ is a true martingale. This is a consequence of Girsanov's theorem [see, e.g., [Karatzas and Shreve, 1991](#), [Revuz and Yor, 1999](#)]. Therefore, to prove [Proposition B.1](#), it comes down to two tasks: showing that D_t is a true martingale, and that $\|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}$ when $h \leq c/(L\sqrt{p(d+p)})$, where c, C are appropriate universal constants.

To show that D_t is a martingale, it suffices to establish Novikov's criterion: $\mathbb{E}(e^{V_h/2}) < \infty$. Since ∇U is L -Lipschitz, one can bound V_h by $L^2 J_h/2$, where $J_h = \int_0^h \|X_t - X_0\|^2 dt$. This prompts us to bound $\mathbb{E}(e^{\lambda J_h})$ for some suitable $\lambda \geq 0$ in [Lemma B.3](#).

We obtain the upper bound $\|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}$ for $h \leq c/(L\sqrt{p(d+p)})$ in [Lemma B.5](#). In the proof of [Lemma B.5](#), we apply Itô's formula, Burkholder-Davis-Gundy (BDG) inequality, and Doob's maximal inequality to obtain what is loosely

$$\|D_h - 1\|_{L^p(\mathbb{P})} \leq C' \sqrt{p} \|V_h\|_{L^p(\mathbb{P})}^{1/2} \leq 2^{-1/2} C' L \sqrt{p} \|J_h\|_{L^p(\mathbb{P})}^{1/2},$$

where C' is universal. This prompts us to study $\|J_h\|_{L^p(\mathbb{P})}$ and obtain an upper bound that is of order $h^2(d+p)$ in [Lemma B.3](#).

To establish the bounds in [Lemma B.3](#), we first control $X_t - X_0$ using stationary reversibility in [Lemma B.2](#), without explicitly solving the Langevin equation.

Once [Proposition B.1](#) is established, we can utilize elementary techniques in the MCMC literature [see, e.g., [Wu et al., 2022](#), proof of Theorem 3] to establish [Proposition 3.2](#).

Attribution within this appendix. The estimate in [Lemma B.2](#) is an elementary consequence of the Lyons–Zheng forward–backward decomposition [[Lyons and Zheng, 1988](#), Section 1, especially equation (1.7)]; see also [Fukushima et al. \[2011, Theorem 5.7.1\]](#). For the Langevin diffusion considered here, the required decomposition follows directly from the stochastic differential equation and stationary reversibility, as shown below. [Lemma B.3](#) then follows from that estimate, Gaussian randomization, and Jensen's inequality. The Gaussian-randomization step is the identity-matrix specialization of the argument used for sub-Gaussian quadratic forms by [Hsu et al. \[2012, Theorem 1 and Remark 2\]](#). [Lemma B.5](#) combines standard results on stochastic exponentials, Novikov's criterion, Girsanov's theorem, stopping of stochastic integrals, Doob's inequality, and the quantitative BDG inequality, with the problem-specific estimate $V_h \leq L^2 J_h/2$. General reverse-Hölder theory for continuous stochastic exponentials is developed by [Kazamaki \[1994\]](#), but the dimension- and moment-explicit calculation needed here is supplied in full. The passage from the path likelihood to the rejection probability uses the standard conditional-likelihood identity and the Metropolis–Hastings meet identity [[Kallenberg, 2021](#), [Tierney, 1998](#), [Billera and Diaconis, 2001](#)]. Thus [Proposition B.1](#) combines these ingredients for the present MALA problem.

B.1 Controlling Langevin increments

The goal of this subsection is to control the magnitude of $X_t - X_0$, particularly the time integral $\int_0^h \|X_t - X_0\|^2 dt$, where X_t satisfies (30).

We begin by fixing the continuous-time framework used throughout this appendix. All random variables are defined on a filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$$

satisfying the usual completeness and right-continuity conditions. A standard d -dimensional Brownian motion relative to this filtration is an adapted continuous process B with $B_0 = 0$ such that, for $0 \leq s < t$, the increment $B_t - B_s$ has law $N_d(0, (t-s)I_d)$ and is independent of $\mathcal{F}_s$. This last independence property will be used explicitly in [Section B.2](#). Basic references for filtered probability spaces, Brownian motion, Itô integration, and stochastic differential equations are [Karatzas and Shreve \[1991, Chapters 3 and 5\]](#), [Revuz and Yor \[1999, Chapters IV and IX\]](#), and [Kallenberg \[2021\]](#).

Because ∇U is globally Lipschitz, for every $x \in \mathbb{R}^d$ the Langevin equation has a unique nonexplosive strong solution $(X_t^x)_{t \geq 0}$ with $X_0^x = x$; see, for example, [Karatzas and Shreve \[1991\]](#),

Chapter 5]. For a bounded measurable function f , define the transition semigroup by

$$S_t f(x) = \mathbb{E}[f(X_t^x)].$$

Now let X_0 be $\mathcal{F}_0$ -measurable and independent of B , with law π , and let $(X_t)_{t \geq 0}$ be the corresponding solution of

$$dX_t = -\nabla U(X_t) dt + \sqrt{2} dB_t, \quad X_0 \sim \pi. \quad (31)$$

The differential notation means the integral identity

$$X_t = X_0 - \int_0^t \nabla U(X_s) ds + \sqrt{2} B_t, \quad t \geq 0. \quad (32)$$

Under the weaker assumptions stated at the beginning of this appendix, the Langevin semigroup is symmetric with respect to $\pi(dx) \propto e^{-U(x)} dx$:

$$\int f S_t g d\pi = \int g S_t f d\pi$$

for bounded measurable f, g and $t \geq 0$; see [Durmus and Moulines \[2017, Section 2\]](#) and [Fukushima et al. \[2011, Chapters 1 and 7\]](#). Since $S_t 1 = 1$, taking $f = 1$ shows that π is invariant. Consequently, the process started from $X_0 \sim \pi$ is stationary and reversible.

Lemma B.2. *For every $v \in \mathbb{R}^d$, $\lambda \in \mathbb{R}$, and $t \geq 0$,*

$$\mathbb{E}[\exp(\lambda \langle v, X_t - X_0 \rangle)] \leq \exp(\lambda^2 t \|v\|^2). \quad (33)$$

Proof. The case $t = 0$ is immediate. Fix $t > 0$. Stationarity and reversibility give

$$(X_s)_{0 \leq s \leq t} \stackrel{d}{=} (X_{t-s})_{0 \leq s \leq t}$$

as random elements of $C([0, t]; \mathbb{R}^d)$. Set

$$Z_t = X_0 - X_t + \int_0^t \nabla U(X_s) ds.$$

Reversing the path interchanges X_0 and X_t , while leaving the time integral unchanged, since

$$\int_0^t \nabla U(X_{t-s}) ds = \int_0^t \nabla U(X_s) ds.$$

The equality of path laws and (32) therefore give

$$Z_t \stackrel{d}{=} X_t - X_0 + \int_0^t \nabla U(X_s) ds = \sqrt{2} B_t \sim N_d(0, 2tI_d).$$

Moreover, subtracting the definition of Z_t from the preceding expression for $\sqrt{2} B_t$ yields

$$X_t - X_0 = \frac{\sqrt{2} B_t - Z_t}{2}.$$

Thus the drift integral cancels when the forward and reversed expressions are subtracted.

By convexity of the exponential function,

$$\begin{aligned}\mathbb{E}[\exp(\lambda \langle v, X_t - X_0 \rangle)] &= \mathbb{E} \left[\exp \left(\frac{\lambda}{2} \langle v, \sqrt{2} B_t - Z_t \rangle \right) \right] \\ &\leq \frac{1}{2} \mathbb{E}[\exp(\sqrt{2} \lambda \langle v, B_t \rangle)] + \frac{1}{2} \mathbb{E}[\exp(-\lambda \langle v, Z_t \rangle)] \\ &= \exp(\lambda^2 t \|v\|^2).\end{aligned}$$

The last equality uses the Gaussian moment-generating formula. No independence between B_t and Z_t is required, and the finite Gaussian exponential moments justify the inequality without any preliminary integrability assumption on its left-hand side. $\square$

The next proof uses a standard Gaussian-randomization argument. In a more general matrix form, it is the argument used by [Hsu et al. \[2012, Theorem 1 and Remark 2\]](#) to pass from linear sub-Gaussian moment bounds to quadratic-form bounds. We include the identity-matrix calculation because the exact constant in (34) is useful below.

Lemma B.3. *For every $t > 0$ and $0 \leq a < 1/(4t)$,*

$$\mathbb{E}[\exp(a \|X_t - X_0\|^2)] \leq (1 - 4at)^{-d/2}. \quad (34)$$

If

$$J_h = \int_0^h \|X_t - X_0\|^2 dt,$$

then, for $0 \leq \lambda < 1/(4h^2)$,

$$\mathbb{E}(e^{\lambda J_h}) \leq (1 - 4\lambda h^2)^{-d/2}. \quad (35)$$

Consequently, for every real $p \geq 1$,

$$\|J_h\|_{L^p(\mathbb{P})} \leq Ch^2(d + p), \quad (36)$$

where C is universal.

Proof. Enlarge the probability space, if necessary, so that it also carries a random vector $G \sim N_d(0, I_d)$ independent of the entire Langevin path $(X_s)_{s \geq 0}$. For a fixed vector $z \in \mathbb{R}^d$, the scalar $\langle G, z \rangle$ is Gaussian with mean zero and variance $\|z\|^2$. The Gaussian moment-generating formula therefore gives

$$\mathbb{E}(e^{\sqrt{2a}\langle G, z \rangle}) = e^{a\|z\|^2}.$$

Apply this identity with the random vector $z = X_t - X_0$. More precisely, conditioning first on the Langevin path gives

$$\mathbb{E}(e^{a\|X_t - X_0\|^2}) = \mathbb{E} \left\{ \mathbb{E} \left[e^{\sqrt{2a}\langle G, X_t - X_0 \rangle} \mid \sigma(X_s : s \geq 0) \right] \right\}.$$

The integrand is nonnegative, so Tonelli's theorem allows us to reverse the order in which the path and G are averaged. Conditional on G , the vector G is fixed, and [Lemma B.2](#), with $v = G$ and $\lambda = \sqrt{2a}$, gives

$$\mathbb{E}(e^{\sqrt{2a}\langle G, X_t - X_0 \rangle} \mid G) \leq e^{2at\|G\|^2}.$$

Consequently,

$$\begin{aligned}\mathbb{E}(e^{a\|X_t - X_0\|^2}) &\leq \mathbb{E}(e^{2at\|G\|^2}) \\ &= (1 - 4at)^{-d/2},\end{aligned}$$

where the last equality is the moment-generating function of a χ_d^2 random variable and requires $4at < 1$. This proves (34).

We next pass from one increment to its time integral. For each fixed sample path, Jensen's inequality for the convex function $x \mapsto e^x$, applied to the uniform probability measure dt/h on $[0, h]$, gives

$$\begin{aligned} e^{\lambda J_h} &= \exp \left(\frac{1}{h} \int_0^h \lambda h \|X_t - X_0\|^2 dt \right) \\ &\leq \frac{1}{h} \int_0^h e^{\lambda h \|X_t - X_0\|^2} dt. \end{aligned}$$

Take expectations and use Tonelli's theorem, followed by (34) with $a = \lambda h$:

$$\begin{aligned} \mathbb{E}(e^{\lambda J_h}) &\leq \frac{1}{h} \int_0^h \mathbb{E}(e^{\lambda h \|X_t - X_0\|^2}) dt \\ &\leq \frac{1}{h} \int_0^h (1 - 4\lambda h t)^{-d/2} dt \\ &\leq (1 - 4\lambda h^2)^{-d/2}. \end{aligned}$$

This proves (35).

It remains to derive the L^p estimate from this exponential-tail bound. Put $\lambda = (8h^2)^{-1}$ in (35). Markov's inequality gives, for every $s \geq 0$,

$$\mathbb{P}(J_h \geq 4dh^2 \log 2 + 8h^2 s) \leq e^{-s}.$$

Define

$$Y = \frac{(J_h - 4dh^2 \log 2)_+}{8h^2}.$$

Then $\mathbb{P}(Y > s) \leq e^{-s}$. For every real $p \geq 1$, the layer-cake formula for moments gives

$$\begin{aligned} \mathbb{E}(Y^p) &= p \int_0^\infty s^{p-1} \mathbb{P}(Y > s) ds \\ &\leq p \int_0^\infty s^{p-1} e^{-s} ds = \Gamma(p+1). \end{aligned}$$

Stirling's bound implies $\Gamma(p+1)^{1/p} \leq Cp$ for all $p \geq 1$. Hence, by the Minkowski inequality,

$$\|J_h\|_{L^p(\mathbb{P})} \leq 4dh^2 \log 2 + 8h^2 \|Y\|_{L^p(\mathbb{P})} \leq Ch^2(d+p),$$

which proves (36). For the tail-integration identity and the Gamma-function estimate, see [Boucheron et al. \[2013, Chapter 2\]](#). $\square$

B.2 Girsanov comparison with frozen drift

We compare two laws on the same path space over $[0, h]$. Under the probability measure $\mathbb{P}$, the path satisfies the exact Langevin equation (31). We shall construct a second measure under which the coordinate process has drift frozen at the initial value $-\nabla U(X_0)$.

We recall the continuous-martingale facts used below. If H_t is a predictable $\mathbb{R}^d$ -valued process with $\int_0^h \|H_t\|^2 dt < \infty$ almost surely, then $\int_0^t \langle H_s, dB_s \rangle$ is a continuous local martingale. Its quadratic variation is

$$\left[\int_0^\cdot \langle H_s, dB_s \rangle \right]_t = \int_0^t \|H_s\|^2 ds.$$

For a continuous local martingale N with $N_0 = 0$, the stochastic exponential is

$$\mathcal{E}(N)_t = \exp(N_t - [N]_t/2).$$

It is always a nonnegative local martingale, but it need not have expectation one. Novikov's criterion states that $\mathbb{E} \exp([N]_h/2) < \infty$ is sufficient for $(\mathcal{E}(N)_t)_{0 \leq t \leq h}$ to be a true mean-one martingale. These statements, together with Itô's formula and the stopping arguments used below, are standard; see [Karatzas and Shreve \[1991, Chapter 3\]](#), [Revuz and Yor \[1999, Chapters IV and VIII\]](#), or [Kallenberg \[2021\]](#).

Define

$$\theta_t = \frac{\nabla U(X_t) - \nabla U(X_0)}{\sqrt{2}}, \quad V_t = \int_0^t \|\theta_s\|^2 ds, \quad M_t = \int_0^t \langle \theta_s, dB_s \rangle.$$

The process θ_t is continuous and adapted, hence predictable. Since X has continuous paths, $\int_0^h \|\theta_s\|^2 ds < \infty$ almost surely on every finite interval, so the Itô integral M_t is well defined as a continuous local $\mathbb{P}$ -martingale. Its quadratic variation is $[M]_t = V_t$. The Lipschitz property of ∇U and [Lemma B.3](#) give

$$V_h \leq \frac{L^2}{2} J_h, \quad \|V_h\|_{L^p(\mathbb{P})} \leq CL^2 h^2 (d+p) \text{ whenever } p \geq 1. \quad (37)$$

Define the stochastic exponential

$$D_t = \exp\left(M_t - \frac{1}{2} V_t\right) = \mathcal{E}(M)_t.$$

The following lemma collects the change-of-measure construction and the endpoint identification that will be used in the proof of [Proposition B.1](#).

Lemma B.4. *Assume that $h > 0$ and $Lh < 1$. Then $(D_t)_{0 \leq t \leq h}$ is a true mean-one $\mathbb{P}$ -martingale, and*

$$\left. \frac{d\mathbb{P}_h^{\text{fr}}}{d\mathbb{P}} \right|_{\mathcal{F}_h} = D_h \quad (38)$$

defines a probability measure $\mathbb{P}_h^{\text{fr}}$ on $(\Omega, \mathcal{F}_h)$. Under $\mathbb{P}_h^{\text{fr}}$, the process

$$W_t = B_t - \int_0^t \theta_s ds, \quad 0 \leq t \leq h,$$

is a Brownian motion relative to $(\mathcal{F}_t)_{0 \leq t \leq h}$, and

$$X_t = X_0 - t \nabla U(X_0) + \sqrt{2} W_t, \quad 0 \leq t \leq h. \quad (39)$$

Moreover, under $\mathbb{P}_h^{\text{fr}}$, $X_0 \sim \pi$, and a version of the conditional law of X_h given $X_0 = x$ is $Q_h(x, \cdot)$. It follows that, for all Borel sets $A, B \subseteq \mathbb{R}^d$,

$$\mathbb{P}_h^{\text{fr}}(X_0 \in A, X_h \in B) = \int_A Q_h(x, B) \pi(dx). \quad (40)$$

Proof. By (35) and (37),

$$\begin{aligned} \mathbb{E} [\exp([M]_h/2)] &= \mathbb{E} [\exp(V_h/2)] \\ &\leq \mathbb{E} [\exp(L^2 J_h/4)] \end{aligned}$$

$$\leq (1 - L^2 h^2)^{-d/2} < \infty,$$

where the last inequality uses $Lh < 1$. Novikov's criterion therefore shows that D is a true mean-one $\mathbb{P}$ -martingale. In particular,

$$\mathbb{E}(D_t \mid \mathcal{F}_s) = D_s, \quad 0 \leq s \leq t \leq h, \quad (41)$$

and (38) defines a probability measure.

Girsanov's theorem gives the asserted Brownian motion W_t ; see Karatzas and Shreve [1991, Section 3.5] or Revuz and Yor [1999, Chapter VIII, Section 1]. Substituting

$$B_t = W_t + \int_0^t \theta_s \, ds$$

into (32) yields

$$\begin{aligned} X_t &= X_0 - \int_0^t \nabla U(X_s) \, ds + \sqrt{2} W_t + \sqrt{2} \int_0^t \theta_s \, ds \\ &= X_0 - t \nabla U(X_0) + \sqrt{2} W_t, \end{aligned}$$

which proves (39).

For every Borel set $A \subseteq \mathbb{R}^d$,

$$\begin{aligned} \mathbb{P}_h^{\text{fr}}(X_0 \in A) &= \mathbb{E} \left(D_h \mathbf{1}_{\{X_0 \in A\}} \right) \\ &= \mathbb{E} \left[\mathbb{E}(D_h \mid \mathcal{F}_0) \mathbf{1}_{\{X_0 \in A\}} \right] \\ &= \mathbb{P}(X_0 \in A) = \pi(A), \end{aligned}$$

where (41) gives $\mathbb{E}[D_h \mid \mathcal{F}_0] = D_0 = 1$.

Under $\mathbb{P}_h^{\text{fr}}$, the increment $W_h - W_0 = W_h$ is independent of $\mathcal{F}_0$ and has law $N_d(0, hI_d)$. Since X_0 is $\mathcal{F}_0$ -measurable, (39) gives, conditionally on $X_0 = x$,

$$X_h = x - h \nabla U(x) + \sqrt{2h} Z, \quad Z \sim N_d(0, I_d).$$

This is precisely the kernel $Q_h(x, \cdot)$. Combining this conditional law with $X_0 \sim \pi$ proves (40). $\square$

We next quantify how close the path likelihood D_h is to one. General reverse-Hölder and moment estimates for continuous stochastic exponentials form a classical theory; see, for example, Kazamaki [1994, Chapters 1 and 3]. We give a direct localized calculation to track explicitly the dependence on d, p, L , and h and to exploit the problem-specific bound $V_h \leq L^2 J_h / 2$.

Lemma B.5. *There are universal constants $c, C > 0$ such that, for every $h > 0$ and every real $p \geq 1$, the conditions*

$$Lh < 1, \quad h \leq \frac{c}{L\sqrt{p(d+p)}} \quad (42)$$

imply

$$\|D_h\|_{L^{2p}(\mathbb{P})} \leq C, \quad \|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}. \quad (43)$$

Proof. Stopping conventions. We use standard process-level stopping identities. If N_t is a continuous local martingale and τ is a stopping time, then, up to indistinguishability,

$$N_t^\tau := N_{t \wedge \tau}, \quad [N^\tau]_t = [N]_{t \wedge \tau}. \quad (44)$$

Likewise, for a predictable integrand H_t ,

$$\left(\int_0^\cdot \langle H_s, dB_s \rangle \right)_{t \wedge \tau} = \int_0^t \mathbf{1}_{\{s \leq \tau\}} \langle H_s, dB_s \rangle. \quad (45)$$

These are identities of stopped processes and therefore hold for all t simultaneously outside a single null set; continuity then permits evaluation at any bounded random time. Purely algebraic identities among continuous processes may be evaluated at $t \wedge \tau$ for the same reason, and inequalities such as $V_{t \wedge \tau} \leq V_t$ follow pathwise from monotonicity. By contrast, an estimate obtained only after taking expectations at each fixed time cannot generally be evaluated at a stopping time without a separate optional-stopping argument. No such step is used here. See [Revuz and Yor \[1999, Chapter IV, Sections 1–2\]](#), [Karatzas and Shreve \[1991, Chapter 3\]](#), or [Protter \[2005, Chapter II\]](#) for stopping of continuous local martingales, stochastic integrals, and quadratic variation.

Step 1: a uniform L^{2p} bound for D_h . For $n \geq 1$, let

$$\tau_n = \inf\{t \in [0, h] : V_t \geq n\}, \quad \inf \emptyset = \infty.$$

Because V is continuous and adapted, τ_n is a stopping time. Set $M_t^{(n)} = M_{t \wedge \tau_n}$. By (44),

$$[M^{(n)}]_t = [M]_{t \wedge \tau_n} = V_{t \wedge \tau_n}.$$

Consequently, the quadratic variation of the continuous local martingale $4pM^{(n)}$ is

$$[4pM^{(n)}]_t = 16p^2 V_{t \wedge \tau_n} \leq 16p^2 n, \quad 0 \leq t \leq h.$$

Consider its stochastic exponential

$$\Lambda_t^{(2p,n)} = \exp\left(4pM_{t \wedge \tau_n} - 8p^2 V_{t \wedge \tau_n}\right).$$

Novikov's criterion applies because

$$\mathbb{E} \left[\exp \left(\frac{1}{2} [4pM^{(n)}]_h \right) \right] \leq e^{8p^2 n} < \infty.$$

Hence $\Lambda^{(2p,n)}$ is a true mean-one martingale, and

$$\mathbb{E} \left[\Lambda_h^{(2p,n)} \right] = 1. \quad (46)$$

The definitions of D and $\Lambda^{(2p,n)}$ give the exact algebraic identity

$$D_{h \wedge \tau_n}^{2p} = [\Lambda_h^{(2p,n)}]^{1/2} e^{(4p^2 - p)V_{h \wedge \tau_n}}.$$

Cauchy–Schwarz, (46), and the inequalities $V_{h \wedge \tau_n} \leq V_h$ and $8p^2 - 2p \geq 0$, valid for $p \geq 1$, yield

$$\begin{aligned} \mathbb{E} \left[D_{h \wedge \tau_n}^{2p} \right] &\leq \left\{ \mathbb{E} [\Lambda_h^{(2p,n)}] \right\}^{1/2} \left\{ \mathbb{E} \left[e^{(8p^2 - 2p)V_{h \wedge \tau_n}} \right] \right\}^{1/2} \\ &\leq \left\{ \mathbb{E} \left[e^{(8p^2 - 2p)V_h} \right] \right\}^{1/2}. \end{aligned}$$

Since $V_h \leq L^2 J_h / 2$, as noted in (37), equation (35) gives

$$\mathbb{E} \left[D_{h \wedge \tau_n}^{2p} \right] \leq (1 - u)^{-d/4}, \quad u = 4p(4p - 1)L^2 h^2,$$

provided $u < 1$. By (42),

$$u \leq 16p^2 L^2 h^2 \leq 16c^2 \frac{p}{d+p} \leq 16c^2.$$

Choose the universal constant $c > 0$ in the statement so that $16c^2 \leq 1/2$. It follows that $0 \leq u \leq 1/2$, and hence

$$\log \left(\frac{1}{1-u} \right) \leq 2u.$$

Therefore

$$\begin{aligned} \log \|D_{h \wedge \tau_n}\|_{L^{2p}(\mathbb{P})} &= \frac{1}{2p} \log \left(\mathbb{E} \left[D_{h \wedge \tau_n}^{2p} \right] \right) \\ &\leq \frac{d}{8p} \log \left(\frac{1}{1-u} \right) \\ &\leq \frac{du}{4p} \\ &\leq 4dp L^2 h^2 \\ &\leq 4c^2 \frac{d}{d+p} \\ &\leq 4c^2. \end{aligned}$$

Thus

$$\|D_{h \wedge \tau_n}\|_{L^{2p}(\mathbb{P})} \leq e^{4c^2} \quad (47)$$

uniformly in n . Finally, $V_h < \infty$ almost surely. For every sample path, once $n > V_h$, one has $\tau_n = \infty$ and hence $D_{h \wedge \tau_n} = D_h$. Fatou's lemma applied to the $(2p)$ -th powers in (47) gives

$$\|D_h\|_{L^{2p}(\mathbb{P})} \leq e^{4c^2}. \quad (48)$$

This proves the first estimate in (43).

Step 2: representation of $D_h - 1$ as a stochastic integral. Let

$$Y_t = M_t - \frac{1}{2}V_t, \quad D_t = e^{Y_t}.$$

Since M is a continuous local $\mathbb{P}$ -martingale and V is continuous and of finite variation, Y is a continuous semimartingale with quadratic variation $[Y]_t = [M]_t = V_t$. Applying Itô's formula for continuous semimartingales [Karatzas and Shreve, 1991, Section 3.3.A, Theorem 3.3] to $y \mapsto e^y$ gives

$$\begin{aligned} dD_t &= D_t dY_t + \frac{1}{2} D_t d[Y]_t \\ &= D_t \left(dM_t - \frac{1}{2} dV_t \right) + \frac{1}{2} D_t dV_t \\ &= D_t dM_t \\ &= D_t \langle \theta_t, dB_t \rangle. \end{aligned}$$

Integrating this identity from zero to t gives the local-martingale representation

$$D_t - 1 = \int_0^t D_s \langle \theta_s, dB_s \rangle. \quad (49)$$

See [Karatzas and Shreve \[1991, Chapter 3\]](#) or [Revuz and Yor \[1999, Chapter IV\]](#) for Itô's formula and the quadratic variation rules used here.

Step 3: application of BDG to an explicitly stopped martingale. For $n \geq 2$, define

$$\sigma_n = \tau_n \wedge \inf\{t \in [0, h] : D_t \geq n\}, \quad \inf \emptyset = \infty.$$

Both V_t and D_t are continuous and adapted, so σ_n is a stopping time. Stopping the process identity (49) and using (45) gives, for every deterministic $t \in [0, h]$,

$$\begin{aligned} R_t^{(n)} &:= D_{t \wedge \sigma_n} - 1 \\ &= \int_0^{t \wedge \sigma_n} D_s \langle \theta_s, dB_s \rangle \\ &= \int_0^t \mathbf{1}_{\{s \leq \sigma_n\}} D_s \langle \theta_s, dB_s \rangle. \end{aligned}$$

Thus $R^{(n)}$ is a square-integrable continuous martingale. Indeed, its quadratic variation is

$$[R^{(n)}]_h = \int_0^h \mathbf{1}_{\{s \leq \sigma_n\}} D_s^2 \|\theta_s\|^2 ds = \int_0^{h \wedge \sigma_n} D_s^2 \|\theta_s\|^2 ds \leq n^2 V_{h \wedge \tau_n} \leq n^3.$$

The possible distinction between $s < \sigma_n$ and $s \leq \sigma_n$ is immaterial for the Lebesgue and Itô integrals.

Apply the upper inequality in [Ren \[2008, Theorem 2, equation \(23\)\]](#), valid for every real $p \geq 1$, to the stopped continuous martingale $R^{(n)}$ at time h . This yields

$$\|R_h^{(n)}\|_{L^p(\mathbb{P})} \leq \left\| \sup_{0 \leq t \leq h} |R_t^{(n)}| \right\|_{L^p(\mathbb{P})} \leq 2\sqrt{2p} \| [R^{(n)}]_h^{1/2} \|_{L^p(\mathbb{P})}. \quad (50)$$

Pathwise,

$$[R^{(n)}]_h^{1/2} \leq \left(\sup_{0 \leq s \leq h} D_s \right) V_h^{1/2}.$$

Hölder's inequality with exponents two and two therefore gives

$$\begin{aligned} \|[R^{(n)}]_h^{1/2}\|_{L^p(\mathbb{P})} &\leq \left\| \sup_{0 \leq s \leq h} D_s \right\|_{L^{2p}(\mathbb{P})} \|V_h^{1/2}\|_{L^{2p}(\mathbb{P})} \\ &= \left\| \sup_{0 \leq s \leq h} D_s \right\|_{L^{2p}(\mathbb{P})} \|V_h\|_{L^p(\mathbb{P})}^{1/2}. \end{aligned} \quad (51)$$

By [Lemma B.4](#), since $Lh < 1$, $(D_t)_{0 \leq t \leq h}$ is a nonnegative true martingale. Conditional Jensen's inequality gives

$$\sup_{0 \leq s \leq h} \|D_s\|_{L^{2p}(\mathbb{P})} = \|D_h\|_{L^{2p}(\mathbb{P})}.$$

Since $2p > 1$, Doob's maximal inequality [[Revuz and Yor, 1999, Chapter II, Theorem 1.7](#)] gives

$$\begin{aligned} \left\| \sup_{0 \leq s \leq h} D_s \right\|_{L^{2p}(\mathbb{P})} &\leq \frac{2p}{2p-1} \sup_{0 \leq s \leq h} \|D_s\|_{L^{2p}(\mathbb{P})} \\ &= \frac{2p}{2p-1} \|D_h\|_{L^{2p}(\mathbb{P})} \leq 2e^{4c^2}, \end{aligned}$$

where the last inequality uses (48) and $2p/(2p-1) \leq 2$ for $p \geq 1$. Moreover, (37) gives

$$\|V_h\|_{L^p(\mathbb{P})}^{1/2} \leq CLh\sqrt{d+p}.$$

Substituting these bounds into (50)–(51) gives, uniformly in n ,

$$\|D_{h \wedge \sigma_n} - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}. \quad (52)$$

To remove the stopping, note that the continuous functions $t \mapsto V_t$ and $t \mapsto D_t$ are bounded on the compact interval $[0, h]$ for every sample path. Hence $\sigma_n > h$ for all sufficiently large n , path by path, and

$$D_{h \wedge \sigma_n} - 1 \longrightarrow D_h - 1 \quad \text{almost surely.}$$

Fatou's lemma applied to (52) proves

$$\|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh\sqrt{p(d+p)}.$$

This is the second estimate in (43) and completes the proof. $\square$

B.3 From path likelihood to MALA rejection

Proof of Proposition B.1. Choose the universal constant c in Proposition B.1 no larger than the constant c in Lemma B.5 and sufficiently small so that the step-size condition $h \leq c/[L\sqrt{p(d+p)}]$ implies $Lh < 1$. This will allow us to apply Lemmas B.4 and B.5.

Let S_h be the exact Langevin transition kernel. Define two endpoint probability measures on $\mathbb{R}^d \times \mathbb{R}^d$:

$$\nu_h(dx, dy) = \pi(dx)S_h(x, dy), \quad \mu_h(dx, dy) = \pi(dx)Q_h(x, dy).$$

The measure ν_h is the law of (X_0, X_h) under the exact path law $\mathbb{P}$. By Lemma B.4, μ_h is the law of (X_0, X_h) under $\mathbb{P}_h^{\text{fr}}$. Moreover, (38) gives

$$\frac{d\mathbb{P}_h^{\text{fr}}}{d\mathbb{P}} = D_h \quad \text{on } \mathcal{F}_h,$$

so $\mu_h \ll \nu_h$. Let

$$F_h = \frac{d\mu_h}{d\nu_h}.$$

We next pass from the full-path likelihood D_h to the endpoint likelihood F_h . By the Doob–Dynkin lemma, $\mathbb{E}[D_h \mid \sigma(X_0, X_h)]$ has a Borel-measurable representation as a function of the two endpoints; this function is a version of F_h . Indeed, for every bounded Borel function $\varphi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$,

$$\begin{aligned} \int \varphi(x, y) \mu_h(dx, dy) &= \mathbb{E}_{\mathbb{P}_h^{\text{fr}}}[\varphi(X_0, X_h)] \\ &= \mathbb{E}[D_h \varphi(X_0, X_h)] \\ &= \mathbb{E}\{\mathbb{E}[D_h \mid \sigma(X_0, X_h)]\varphi(X_0, X_h)\}. \end{aligned}$$

This is exactly the defining Radon–Nikodym identity. Thus

$$F_h(X_0, X_h) = \mathbb{E}[D_h \mid \sigma(X_0, X_h)] \quad \mathbb{P}\text{-almost surely.} \quad (53)$$

This is the usual principle that the likelihood ratio of an observed statistic is the conditional expectation of the full-data likelihood ratio; see Kallenberg [2021] for conditional expectation, disintegration, and regular conditional distributions.

Subtracting one in (53) and using conditional Jensen's inequality gives the L^p -contraction explicitly:

$$\begin{aligned}\|F_h - 1\|_{L^p(\nu_h)}^p &= \mathbb{E} |\mathbb{E}[D_h - 1 \mid \sigma(X_0, X_h)]|^p \\ &\leq \mathbb{E} \{ \mathbb{E} [|D_h - 1|^p \mid \sigma(X_0, X_h)] \} \\ &= \|D_h - 1\|_{L^p(\mathbb{P})}^p.\end{aligned}$$

Hence

$$\|F_h - 1\|_{L^p(\nu_h)} \leq \|D_h - 1\|_{L^p(\mathbb{P})}. \quad (54)$$

We also need the conditional form of $\mu_h = F_h \nu_h$. Evaluating this measure identity on a rectangle $A \times B$ gives

$$\int_A Q_h(x, B) \pi(dx) = \int_A \int_B F_h(x, y) S_h(x, dy) \pi(dx). \quad (55)$$

Fix a countable π -system $\mathcal{C}$ that generates the Borel σ -field on $\mathbb{R}^d$, for example, the finite intersections of open balls with rational centers and radii, together with $\mathbb{R}^d$. For each $B \in \mathcal{C}$, equality (55) for every Borel A implies equality of the two integrands for π -almost every x , i.e.,

$$Q_h(x, B) = \int_B F_h(x, y) S_h(x, dy).$$

Since $\mathcal{C}$ is countable, there is one π -null set outside which this holds simultaneously for every $B \in \mathcal{C}$. For each fixed x outside that null set, both sides are measures in B , and the monotone-class theorem extends their equality from $\mathcal{C}$ to all Borel sets. Thus, for π -almost every x ,

$$Q_h(x, dy) = F_h(x, y) S_h(x, dy). \quad (56)$$

By reversibility, ν_h is symmetric. Hence the transpose measure $\mu_h^\top(dx, dy) = \mu_h(dy, dx)$ has density $F_h(y, x)$ with respect to ν_h . Since the target and proposal densities are strictly positive, the Radon–Nikodym chain rule gives

$$F_h(y, x) = \frac{e^{-U(y)} q_h(y, x)}{e^{-U(x)} q_h(x, y)} F_h(x, y) \quad \nu_h\text{-almost everywhere.}$$

The definition of the Metropolis–Hastings acceptance probability therefore implies

$$F_h(x, y) \alpha_h(x, y) = \min\{F_h(x, y), F_h(y, x)\} \quad \nu_h\text{-almost everywhere.}$$

This is the usual Metropolis–Hastings minimum identity [Tierney, 1998, Billera and Diaconis, 2001].

Because $\nu_h(dx, dy) = \pi(dx) S_h(x, dy)$, Tonelli's theorem shows that the preceding identity holds for $S_h(x, \cdot)$ -almost every y , for π -almost every x . Using (56), we obtain, for π -almost every x ,

$$\begin{aligned}r_h(x) &= 1 - \int \alpha_h(x, y) Q_h(x, dy) \\ &= 1 - \int \alpha_h(x, y) F_h(x, y) S_h(x, dy) \\ &= \int [1 - \min\{F_h(x, y), F_h(y, x)\}] S_h(x, dy) \\ &\leq \int (|F_h(x, y) - 1| + |F_h(y, x) - 1|) S_h(x, dy).\end{aligned}$$

Define

$$A_h(x) = \int |F_h(x, y) - 1| S_h(x, dy), \quad B_h(x) = \int |F_h(y, x) - 1| S_h(x, dy).$$

Then $r_h \leq A_h + B_h$. Jensen's inequality for the probability measure $S_h(x, dy)$ gives

$$\begin{aligned} \|A_h\|_{L^p(\pi)}^p &\leq \int |F_h(x, y) - 1|^p \nu_h(dx, dy), \\ \|B_h\|_{L^p(\pi)}^p &\leq \int |F_h(y, x) - 1|^p \nu_h(dx, dy). \end{aligned}$$

The first right-hand side is $\|F_h - 1\|_{L^p(\nu_h)}^p$; the second is the same because ν_h is symmetric. Minkowski's inequality therefore yields

$$\|r_h\|_{L^p(\pi)} \leq \|A_h\|_{L^p(\pi)} + \|B_h\|_{L^p(\pi)} \leq 2 \|F_h - 1\|_{L^p(\nu_h)}. \quad (57)$$

Combining (57), (54), and Lemma B.5 gives

$$\|r_h\|_{L^p(\pi)} \leq 2 \|D_h - 1\|_{L^p(\mathbb{P})} \leq CLh \sqrt{p(d+p)},$$

after absorbing the factor two into the universal constant. This is (29). $\square$

B.4 Proving Proposition 3.2

Proof. We begin with the first of the two assertions. Let c be the step-size constant in Proposition B.1, and choose $0 < c_r \leq \min\{c, 1\}$. Fix $p \geq 1$ and t satisfying (6). Then Proposition B.1 applies to every $h \in [t/2, t]$, and

$$Lt \leq \frac{c_r}{\sqrt{p(d+p)}} \leq \frac{1}{\sqrt{2}} < 2.$$

Define the rejection probability averaged over $[t/2, t]$ by

$$\bar{r}_t(x) = \frac{2}{t} \int_{t/2}^t r_h(x) dh.$$

Minkowski's integral inequality and Proposition B.1 give

$$\begin{aligned} \|\bar{r}_t\|_{L^p(\pi)} &\leq \frac{2}{t} \int_{t/2}^t \|r_h\|_{L^p(\pi)} dh \\ &\leq CL \sqrt{p(d+p)} \frac{2}{t} \int_{t/2}^t h dh = \frac{3C}{4} Lt \sqrt{p(d+p)}. \end{aligned}$$

Set

$$G_{t,p} = \{x : \bar{r}_t(x) \leq 1/3\}.$$

Markov's inequality yields

$$\pi(G_{t,p}^c) \leq \left(\frac{9C}{4} Lt \sqrt{p(d+p)} \right)^p,$$

which is (7) after renaming the universal constant.

We next compare nearby proposal laws. Because U is convex and has L -Lipschitz gradient, the Baillon–Haddad cocoercivity inequality states that [Nesterov, 2004, Theorem 2.1.5]

$$\langle \nabla U(x) - \nabla U(y), x - y \rangle \geq \frac{1}{L} \|\nabla U(x) - \nabla U(y)\|^2.$$

See also [Bauschke and Combettes \[2010\]](#). Consequently, for $h \leq t$,

$$\begin{aligned} & \|x - h\nabla U(x) - y + h\nabla U(y)\|^2 \\ &= \|x - y\|^2 - 2h \langle x - y, \nabla U(x) - \nabla U(y) \rangle + h^2 \|\nabla U(x) - \nabla U(y)\|^2 \\ &\leq \|x - y\|^2. \end{aligned}$$

Note that we have used $h \leq t \leq 2/L$. Thus the proposal mean map $x \mapsto x - h\nabla U(x)$ is nonexpansive. Two Gaussian laws with covariance $2hI_d$ and means μ_x, μ_y have Kullback–Leibler divergence $\|\mu_x - \mu_y\|^2/(4h)$. Pinsker’s inequality [[Tsybakov, 2009](#), Lemma 2.5] therefore gives

$$\text{TV}(Q_h(x, \cdot), Q_h(y, \cdot)) \leq \frac{\|x - y\|}{\sqrt{8h}} \leq \frac{\|x - y\|}{2\sqrt{t}}, \quad h \in [t/2, t]. \quad (58)$$

For fixed x , write

$$R_{h,x}(\mathrm{d}y) = [1 - \alpha_h(x, y)] Q_h(x, \mathrm{d}y).$$

This measure has total mass $r_h(x)$, and, because $Q_h(x, \cdot)$ has a Lebesgue density, it gives no mass to $\{x\}$. Hence

$$P_h(x, \cdot) - Q_h(x, \cdot) = r_h(x)\delta_x - R_{h,x}$$

is the Jordan decomposition into mutually singular positive and negative parts. This proves

$$\text{TV}(P_h(x, \cdot), Q_h(x, \cdot)) = r_h(x). \quad (59)$$

Total variation is convex under mixing: for probability kernels M_h, N_h and a probability measure ρ ,

$$\text{TV}\left(\int M_h \mathrm{d}\rho, \int N_h \mathrm{d}\rho\right) \leq \int \text{TV}(M_h, N_h) \mathrm{d}\rho.$$

Applying this fact three times, together with the triangle inequality, (59), and (58), gives, for $x, y \in G_{t,p}$ with $\|x - y\| \leq \sqrt{t}/16$,

$$\begin{aligned} \text{TV}(\mathcal{K}_t(x, \cdot), \mathcal{K}_t(y, \cdot)) &\leq \bar{r}_t(x) + \frac{2}{t} \int_{t/2}^t \text{TV}(Q_h(x, \cdot), Q_h(y, \cdot)) \mathrm{d}h + \bar{r}_t(y) \\ &\leq \frac{1}{3} + \frac{1}{32} + \frac{1}{3} = \frac{67}{96} < \frac{3}{4}. \end{aligned}$$

This proves the first assertion of [Proposition 3.2](#). We note that this type of argument is common in the MCMC literature; see, e.g., the proof of Theorem 3 in [Wu et al. \[2022\]](#).

It remains to prove the final assertion. Let $Z \sim N_d(0, I_d)$, and $Y = x - h\nabla U(x) + \sqrt{2h}Z$. We first derive the log-ratio identity used below. Since, for $a, b \in \mathbb{R}^d$,

$$q_h(a, b) = (4\pi h)^{-d/2} \exp\left[-\frac{\|b - a + h\nabla U(a)\|^2}{4h}\right],$$

one has

$$\begin{aligned} \log \frac{e^{-U(Y)} q_h(Y, x)}{e^{-U(x)} q_h(x, Y)} &= U(x) - U(Y) \\ &\quad + \frac{1}{4h} \left(\|Y - x + h\nabla U(x)\|^2 - \|x - Y + h\nabla U(Y)\|^2 \right). \end{aligned}$$

Expanding the two squared norms and collecting the gradient and Gaussian terms gives

$$\begin{aligned} \log \frac{e^{-U(Y)} q_h(Y, x)}{e^{-U(x)} q_h(x, Y)} &= U(x) - U(Y) - \langle \nabla U(Y), x - Y \rangle - \frac{h}{4} \|\nabla U(Y) - \nabla U(x)\|^2 \\ &\quad - \sqrt{\frac{h}{2}} \langle Z, \nabla U(Y) - \nabla U(x) \rangle. \end{aligned} \quad (60)$$

The Baillon–Haddad inequality [Nesterov, 2004, Theorem 2.1.5] gives

$$U(x) - U(Y) - \langle \nabla U(Y), x - Y \rangle \geq \frac{1}{2L} \|\nabla U(Y) - \nabla U(x)\|^2.$$

If $h \leq 1/L$, the right-hand side of (60) is therefore at least

$$\frac{1}{4L} \|\nabla U(Y) - \nabla U(x)\|^2 - \sqrt{\frac{h}{2}} \|Z\| \|\nabla U(Y) - \nabla U(x)\| \geq -\frac{Lh}{2} \|Z\|^2.$$

It follows that

$$\alpha_h(x, Y) \geq e^{-Lh\|Z\|^2/2}, \quad \mathbb{E}[\alpha_h(x, Y)] \geq (1 + Lh)^{-d/2} \geq e^{-Lhd/2}.$$

Let $t' \in (0, 1/(2Ld)]$. Then every $h \in [t'/2, t']$ has rejection probability at most $1 - e^{-1/4}$, uniformly in x . Combining two such rejection bounds with (58), we obtain, whenever $\|x - y\| \leq \sqrt{t'}/16$,

$$\text{TV}(\mathcal{K}_{t'}(x, \cdot), \mathcal{K}_{t'}(y, \cdot)) \leq 2(1 - e^{-1/4}) + \frac{1}{32} < \frac{3}{4}.$$

This proves the final assertion of Proposition 3.2 and completes the proof. $\square$

C Proof of Proposition 3.3

We repeatedly use the following notation for the standard Gaussian distribution:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad \Phi(x) = \int_{-\infty}^x \varphi(v) \, dv, \quad \bar{\Phi}(x) = 1 - \Phi(x).$$

C.1 Some basic properties of Gaussian distribution functions

The Gaussian enlargement inequality (8), justified above under the first-order assumptions, states that

$$\pi(A^r) \geq \Phi(\Phi^{-1}(\pi(A)) + \sqrt{m}r).$$

We analyze the map $(q, s) \mapsto \Phi(\Phi^{-1}(q) + s)$.

We begin with a Mills-ratio bound.

Lemma C.1. *Let $a \geq 0$. Then*

$$\frac{\varphi(a)}{1+a} \leq \bar{\Phi}(a) \leq \frac{2\varphi(a)}{1+a}. \quad (61)$$

Moreover,

$$\log \frac{1}{\bar{\Phi}(a)} \leq (1+a)^2. \quad (62)$$

Proof. Let

$$g(a) = \bar{\Phi}(a) - \frac{\varphi(a)}{1+a}.$$

Then $g(a) \rightarrow 0$ as $a \rightarrow \infty$, and direct differentiation gives

$$g'(a) = -\frac{a\varphi(a)}{(1+a)^2} \leq 0.$$

Thus $g(a) \geq 0$, proving the lower bound in (61). Similarly, for

$$f(a) = \frac{2\varphi(a)}{1+a} - \bar{\Phi}(a),$$

we have $f(a) \rightarrow 0$ as $a \rightarrow \infty$ and

$$f'(a) = -\frac{(1+a^2)\varphi(a)}{(1+a)^2} < 0.$$

Hence $f(a) \geq 0$, proving the upper bound. Classical sharper Mills bounds for $a > 0$ go back at least to [Gordon \[1941\]](#).

The lower bound in (61) implies

$$\log \frac{1}{\bar{\Phi}(a)} \leq \frac{a^2}{2} + \log(1+a) + \frac{1}{2} \log(2\pi).$$

The difference between $(1+a)^2$ and the right-hand side is positive at $a = 0$, and its derivative is

$$2+a - \frac{1}{1+a} > 0.$$

This proves (62). □

Lemma C.2. *Let $q \in (0, 1/2]$ and $s \geq 0$. Then*

$$\Phi(\Phi^{-1}(q) + s) - q \geq \frac{q}{4} \min \left\{ 1, s\sqrt{\log(1/q)} \right\}.$$

Proof. Write $\Phi^{-1}(q) = -a$, so $a \geq 0$ and $q = \bar{\Phi}(a)$. Then

$$\Phi(-a+s) - \Phi(-a) = \int_0^s \varphi(a-v) dv.$$

For $0 \leq v \leq \min\{s, 1\}$,

$$\frac{\varphi(a-v)}{\varphi(a)} = e^{av-v^2/2} \geq e^{-1/2}.$$

By the upper Mills bound in (61), $\varphi(a) \geq q(1+a)/2$. Hence

$$\begin{aligned} \Phi(-a+s) - \Phi(-a) &\geq \frac{e^{-1/2}}{2} q(1+a) \min\{s, 1\} \\ &\geq \frac{e^{-1/2}}{2} q \min\{1, s(1+a)\} \\ &\geq \frac{q}{4} \min \left\{ 1, s\sqrt{\log(1/q)} \right\}. \end{aligned}$$

The second line uses $(1+a) \min\{s, 1\} \geq \min\{1, s(1+a)\}$; the third uses (62) and $e^{-1/2}/2 > 1/4$. □

C.2 Proving Proposition 3.3

Proof. If $r = 0$, the right-hand side of (9) is zero, so the conclusion is immediate. Assume $r > 0$ and, without loss of generality, that $\pi(A) = q$. Since $\text{dist}(A, B) \geq r$, the open enlargement A^r is disjoint from B , and hence $A^r \subseteq A \cup C$. Combining (8) and Lemma C.2 gives

$$\begin{aligned}\pi(C) &\geq \pi(A^r) - \pi(A) \\ &\geq \Phi(\Phi^{-1}(q) + \sqrt{m}r) - q \\ &\geq \frac{q}{4} \min \left\{ 1, r\sqrt{m \log(1/q)} \right\}.\end{aligned}$$

This is exactly (9). □

D Proof of Proposition 3.4

D.1 A generic conductance bound

We begin with a generic one-step flow bound for kernels that satisfy a close-coupling condition under the Gaussian isoperimetric inequality in Proposition 3.3. Combining close coupling with an isoperimetric inequality to lower-bound one-step flows, and hence conductance, is a standard technique. Related defective- or truncated-conductance arguments appear in Dwivedi et al. [2019], Wu et al. [2022], Chen and Gatmiry [2023], Liu and Tong [2026].

Lemma D.1. *Let K be π -reversible, let $t > 0$, and let $G \subseteq \mathbb{R}^d$ satisfy*

$$x, y \in G, \quad \|x - y\| \leq \frac{\sqrt{t}}{16} \implies \text{TV}(K(x, \cdot), K(y, \cdot)) \leq \frac{3}{4}.$$

Let S be a measurable set such that $0 < \pi(S) \leq 1/2$, and suppose that

$$\pi(G^c) \leq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt \log[1/\pi(S)]} \right\}. \quad (63)$$

Then

$$\mathcal{J}_K(S, S^c) \geq 2^{-13} \pi(S) \min \left\{ 1, \sqrt{mt \log[1/\pi(S)]} \right\}. \quad (64)$$

Proof. Fix a measurable S with $0 < \pi(S) \leq 1/2$. Let

$$\beta = \min \left\{ 1, \sqrt{mt \log[1/\pi(S)]} \right\}.$$

Suppose, for contradiction, that

$$\mathcal{J}_K(S, S^c) < 2^{-13} \beta \pi(S). \quad (65)$$

Define

$$\begin{aligned}S_1 &= \{x \in S \cap G : K(x, S^c) < 1/16\}, \\ S_2 &= \{y \in S^c \cap G : K(y, S) < 1/16\}.\end{aligned}$$

For $x \in S_1$ and $y \in S_2$,

$$\text{TV}(K(x, \cdot), K(y, \cdot)) \geq K(x, S) - K(y, S) > \frac{7}{8}.$$

The close-coupling condition therefore implies

$$\text{dist}(S_1, S_2) \geq \frac{\sqrt{t}}{16}. \quad (66)$$

Markov's inequality and reversibility give

$$\pi(S_1^c \cap S \cap G) \leq 16\mathcal{J}_K(S, S^c), \quad \pi(S_2^c \cap S^c \cap G) \leq 16\mathcal{J}_K(S, S^c). \quad (67)$$

Thus,

$$\begin{aligned} \pi(S_1) &\geq \pi(S) - \pi(S \cap G^c) - 16\mathcal{J}_K(S, S^c), \\ \pi(S_2) &\geq 1 - \pi(S) - \pi(S^c \cap G^c) - 16\mathcal{J}_K(S, S^c). \end{aligned}$$

By (63) and the assumed upper bound (65),

$$\pi(S_1) > \pi(S) - 17 \cdot 2^{-13} \beta \pi(S), \quad \pi(S_2) > 1 - \pi(S) - 17 \cdot 2^{-13} \beta \pi(S).$$

Since $0 < \beta \leq 1$, $17 \cdot 2^{-13} < 1/2$, and $1 - \pi(S) \geq \pi(S)$, it follows that

$$\pi(S_1) \geq \frac{\pi(S)}{2}, \quad \pi(S_2) \geq \frac{\pi(S)}{2}. \quad (68)$$

Set $q' = \min\{\pi(S_1), \pi(S_2)\}$. Because $S_1 \subseteq S$,

$$\frac{\pi(S)}{2} \leq q' \leq \pi(S) \leq \frac{1}{2}.$$

Apply Proposition 3.3 to S_1, S_2 , and their complement. Since $q' \leq \pi(S)$, one has $\log(1/q') \geq \log[1/\pi(S)]$. Using (66), (68), and $\min\{1, x/16\} \geq \min\{1, x\}/16$, we obtain

$$\begin{aligned} \pi((S_1 \cup S_2)^c) &\geq \frac{q'}{4} \min\left\{1, \frac{1}{16} \sqrt{mt \log(1/q')}\right\} \\ &\geq \frac{\pi(S)}{8} \cdot \frac{1}{16} \min\left\{1, \sqrt{mt \log[1/\pi(S)]}\right\} = \frac{\beta \pi(S)}{128}. \end{aligned} \quad (69)$$

On the other hand, by (67), (63), and (65),

$$\begin{aligned} \pi((S_1 \cup S_2)^c) &= \pi(G^c) + \pi(G \cap S \cap S_1^c) + \pi(G \cap S^c \cap S_2^c) \\ &\leq \pi(G^c) + 32\mathcal{J}_K(S, S^c) \\ &< 33 \cdot 2^{-13} \beta \pi(S) < \frac{\beta \pi(S)}{128}, \end{aligned}$$

contradicting (69). Therefore the assumed upper bound (65) is impossible, and (64) holds. $\square$

D.2 A technical lemma

The proof of Proposition 3.4 via Lemma D.1 requires one additional technical lemma.

Lemma D.2. *Let c_r, C_r be the constants in Proposition 3.2. Fix a number $b_0 > 0$ satisfying*

$$b_0 \leq \min\left\{\frac{1}{2}, c_r, \frac{1}{16C_r}\right\}. \quad (70)$$

Let $p \geq p_\star := A_0[1 + \log(d+1) + \log \kappa]$, $\theta \in (0, 1]$, and

$$t = \frac{\theta b_0}{L\sqrt{p(d+p)}},$$

where A_0 satisfies

$$A_0 \geq \frac{13 \log 2 + \frac{1}{2} \log(2/b_0)}{\log 16 - \frac{1}{2}}. \quad (71)$$

Then, for every $u \in [\log 2, p/2]$,

$$\left(\frac{\theta}{16}\right)^p \leq 2^{-13} e^{-u} \sqrt{mtu}. \quad (72)$$

Moreover, $mtu \leq b_0/2 < 1$ throughout this range.

Proof. Since $b_0 \leq 1/2$, the bound (71) implies $A_0 \geq 1$. Consequently,

$$p \geq p_\star \geq 1 + \log 2 > 2 \log 2 > 1,$$

so $[\log 2, p/2]$ is nonempty. The last assertion follows directly from $m/L = 1/\kappa$:

$$mtu = \frac{\theta b_0 u}{\kappa \sqrt{p(d+p)}} \leq \frac{\theta b_0}{2\kappa} \sqrt{\frac{p}{d+p}} \leq \frac{b_0}{2}.$$

We focus on proving (72). Since $u \mapsto e^{-u} \sqrt{u}$ decreases on $[1/2, \infty)$, the right-hand side of (72) is smallest at $u = p/2$. At that endpoint it equals

$$2^{-13} e^{-p/2} \left(\frac{\theta b_0}{2\kappa}\right)^{1/2} \left(\frac{p}{d+p}\right)^{1/4}.$$

Dividing the desired inequality at $u = p/2$ by $\theta^{1/2}$ shows that its left-hand side contains the factor $\theta^{p-1/2}$. Since $p \geq 1$ and $0 < \theta \leq 1$, this factor is at most one. It is therefore sufficient that

$$p \left(\log 16 - \frac{1}{2}\right) \geq 13 \log 2 + \frac{1}{2} \log \frac{2\kappa}{b_0} + \frac{1}{4} \log \frac{d+p}{p}. \quad (73)$$

Since $p \geq 1$,

$$\log \frac{d+p}{p} \leq \log(d+1).$$

The right-hand side of (73) is therefore at most

$$13 \log 2 + \frac{1}{2} \log \frac{2}{b_0} + \frac{1}{2} \log \kappa + \frac{1}{4} \log(d+1).$$

By (71) and $b_0 \leq 1/2$,

$$A_0 \left(\log 16 - \frac{1}{2}\right) \geq 13 \log 2 + \frac{1}{2} \log(2/b_0) \geq 14 \log 2 > \frac{1}{2}.$$

Since $\log(d+1)$ and $\log \kappa$ are nonnegative, these bounds and $p \geq p_\star$ give

$$\begin{aligned} p \left(\log 16 - \frac{1}{2}\right) &\geq A_0 \left(\log 16 - \frac{1}{2}\right) [1 + \log(d+1) + \log \kappa] \\ &\geq 13 \log 2 + \frac{1}{2} \log(2/b_0) + \frac{1}{4} \log(d+1) + \frac{1}{2} \log \kappa. \end{aligned}$$

This proves (73), uniformly in all parameters. $\square$

D.3 Proving Proposition 3.4

Proof. We begin with the first assertion. Let A_0 and b_0 be universal constants satisfying (70) and (71). Let $p \geq p_\star := A_0[1 + \log(d+1) + \log \kappa]$, $\theta \in (0, 1]$, and

$$t = \frac{\theta b_0}{L\sqrt{p(d+p)}}.$$

Let $S \subseteq \mathbb{R}^d$ be measurable with $e^{-p/2} \leq \pi(S) \leq 1/2$. By Lemma D.2, since $\log[1/\pi(S)] \in [\log 2, p/2]$,

$$\left(\frac{\theta}{16}\right)^p \leq 2^{-13}\pi(S)\sqrt{mt \log[1/\pi(S)]} = 2^{-13}\pi(S) \min \left\{1, \sqrt{mt \log[1/\pi(S)]}\right\}.$$

The equality follows from the final assertion of Lemma D.2, namely, $mt \log[1/\pi(S)] \leq 1$. Because $b_0 \leq c_r$ and $b_0 \leq 1/(16C_r)$, the first assertion of Proposition 3.2 yields a measurable set $G_{t,p} \subseteq \mathbb{R}^d$ such that

$$\pi(G_{t,p}^c) \leq \left[C_r L \frac{\theta b_0}{L\sqrt{p(d+p)}} \sqrt{p(d+p)} \right]^p \leq \left(\frac{\theta}{16}\right)^p,$$

and

$$x, y \in G_{t,p}, \quad \|x - y\| \leq \frac{\sqrt{t}}{16} \implies \text{TV}(\mathcal{K}_t(x, \cdot), \mathcal{K}_t(y, \cdot)) \leq \frac{3}{4}.$$

Then, by Lemma D.1,

$$\mathcal{J}_{\mathcal{K}_t}(S, S^c) \geq 2^{-13}\pi(S) \min \left\{1, \sqrt{mt \log[1/\pi(S)]}\right\} = 2^{-13}\pi(S)\sqrt{mt \log[1/\pi(S)]}.$$

This proves the first assertion.

For the second assertion, let $t' \in (0, 1/(2Ld)]$ and let $S \subseteq \mathbb{R}^d$ be measurable with $0 < \pi(S) \leq 1/2$. By the second assertion of Proposition 3.2,

$$x, y \in \mathbb{R}^d, \quad \|x - y\| \leq \frac{\sqrt{t'}}{16} \implies \text{TV}(\mathcal{K}_{t'}(x, \cdot), \mathcal{K}_{t'}(y, \cdot)) \leq \frac{3}{4}.$$

Applying Lemma D.1 with $G = \mathbb{R}^d$ yields

$$\mathcal{J}_{\mathcal{K}_{t'}}(S, S^c) \geq 2^{-13}\pi(S) \min \left\{1, \sqrt{mt' \log[1/\pi(S)]}\right\}.$$

This proves the second assertion. □

E Proof of Lemma 3.5

The result can also be derived from the general Cheeger inequality of Chen and Wang [2000]. The proof below is direct and self-contained; we include the details because the precise dependence on β_j and γ_j is needed later.

Proof. For $j = 1, \dots, N$, define the symmetric measure

$$J_j(dx, dy) = \varpi(dx)K_j(x, dy).$$

The measure J_j is symmetric because K_j is ϖ -reversible. Both of its marginals equal ϖ .

We first prove

$$\sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g, g) \geq \frac{\|g\|_{L^2(\varpi)}^2}{2 \sum_{j=1}^N \beta_j^2 / \gamma_j} \quad (74)$$

for a nonnegative function $g \in L^2(\varpi)$ such that $\varpi(g > 0) \leq 1/2$. For $t \geq 0$, let

$$A_t = \{x : g(x)^2 > t\}.$$

Then $\varpi(A_t) \leq 1/2$, so the assumed fractional boundary inequality gives

$$\varpi(A_t) \leq \sum_{j=1}^N \beta_j \mathcal{J}_{K_j}(A_t, A_t^c).$$

Integrating this inequality over $t \geq 0$ and using the layer-cake identity yields

$$\begin{aligned} \|g\|_{L^2(\varpi)}^2 &= \int_0^\infty \varpi(A_t) dt \\ &\leq \sum_{j=1}^N \beta_j \int_0^\infty \mathcal{J}_{K_j}(A_t, A_t^c) dt. \end{aligned}$$

For each j , symmetry of J_j gives the coarea identity

$$\int_0^\infty \mathcal{J}_{K_j}(A_t, A_t^c) dt = \frac{1}{2} \iint |g(x)^2 - g(y)^2| J_j(dx, dy).$$

Indeed, for fixed x, y ,

$$\int_0^\infty \mathbf{1}_{\{g(x)^2 > t \geq g(y)^2\}} dt = [g(x)^2 - g(y)^2]_+,$$

and symmetry changes the integral of the positive part into one half of the integral of the absolute difference. Consequently,

$$\|g\|_{L^2(\varpi)}^2 \leq \frac{1}{2} \sum_{j=1}^N \beta_j \iint |g(x) - g(y)| |g(x) + g(y)| J_j(dx, dy).$$

Apply Cauchy–Schwarz simultaneously over the component index j and the pair (x, y) :

$$\begin{aligned} \|g\|_{L^2(\varpi)}^2 &\leq \frac{1}{2} \left\{ \sum_{j=1}^N \gamma_j \iint [g(x) - g(y)]^2 J_j(dx, dy) \right\}^{1/2} \\ &\quad \times \left\{ \sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} \iint [g(x) + g(y)]^2 J_j(dx, dy) \right\}^{1/2}. \end{aligned}$$

The first square root equals

$$\left[2 \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g, g) \right]^{1/2}.$$

For the second square root, the inequality

$$[g(x) + g(y)]^2 \leq 2g(x)^2 + 2g(y)^2$$

and the fact that both marginals of J_j equal ϖ give

$$\iint [g(x) + g(y)]^2 J_j(dx, dy) \leq 4 \|g\|_{L^2(\varpi)}^2.$$

It follows that

$$\|g\|_{L^2(\varpi)}^2 \leq \left[2 \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g, g) \right]^{1/2} \|g\|_{L^2(\varpi)} \left(\sum_{j=1}^N \frac{\beta_j^2}{\gamma_j} \right)^{1/2}.$$

If $g = 0$ in $L^2(\varpi)$, there is nothing to prove. Otherwise, cancelling one factor of $\|g\|_{L^2(\varpi)}$ and squaring gives (74).

Now let $f \in L^2(\varpi)$, and let c be a median of f ; thus

$$\varpi(f \geq c) \geq \frac{1}{2}, \quad \varpi(f \leq c) \geq \frac{1}{2}.$$

Define the positive and negative parts of $f - c$:

$$g_+ = (f - c)_+, \quad g_- = (f - c)_-.$$

Then

$$\varpi(g_+ > 0) \leq \frac{1}{2}, \quad \varpi(g_- > 0) \leq \frac{1}{2}.$$

Moreover, for every x, y ,

$$[f(x) - f(y)]^2 \geq [g_+(x) - g_+(y)]^2 + [g_-(x) - g_-(y)]^2.$$

To see this, write

$$f - c = g_+ - g_-.$$

The two differences

$$g_+(x) - g_+(y) \quad \text{and} \quad g_-(x) - g_-(y)$$

have opposite signs, so the cross term in the square of their difference is nonnegative.

Therefore, by (74),

$$\begin{aligned} \mathcal{E}_P(f, f) &\geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(f, f) \\ &\geq \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g_+, g_+) + \sum_{j=1}^N \gamma_j \mathcal{E}_{K_j}(g_-, g_-) \\ &\geq \frac{\|g_+\|_{L^2(\varpi)}^2 + \|g_-\|_{L^2(\varpi)}^2}{2 \sum_{j=1}^N \beta_j^2 / \gamma_j}. \end{aligned}$$

Since g_+ and g_- have disjoint supports and $f - c = g_+ - g_-$,

$$\|g_+\|_{L^2(\varpi)}^2 + \|g_-\|_{L^2(\varpi)}^2 = \|f - c\|_{L^2(\varpi)}^2.$$

The mean minimizes $a \mapsto \|f - a\|_{L^2(\varpi)}^2$, so

$$\|f - c\|_{L^2(\varpi)}^2 \geq \text{Var}_{\varpi}(f).$$

We conclude that

$$\mathcal{E}_P(f, f) \geq \frac{\text{Var}_{\varpi}(f)}{2 \sum_{j=1}^N \beta_j^2 / \gamma_j}.$$

Taking the infimum over all nonconstant $f \in L^2(\varpi)$ proves the claim. $\square$

F Multiscale arithmetic

This appendix verifies the summation estimate used in [Section 4](#). The constants $b_0 \leq 1/2$ and $A_0 \geq 1$ are the universal choices made at the beginning of the proof of [Theorem 2.1](#) in [Section 4](#), and $p_\star = A_0[1 + \log(d+1) + \log \kappa]$.

Lemma F.1. *Assume $p_\star < d$, and define p_j, J, t_j, θ by (10)–(12). With*

$$\gamma_j = \frac{t_j}{2H}, \quad \phi_0^2 = 2^{-26} m t_0 \log 2, \quad \phi_j^2 = 2^{-29} m t_j p_j \quad (j \geq 1),$$

there is a universal $C \in (0, \infty)$ such that

$$\sum_{j=0}^J \frac{1}{\gamma_j \phi_j^2} \leq C \frac{H L^2 p_\star (d + p_\star)}{m \theta^2 b_0^2}.$$

Proof. For the first component,

$$\frac{1}{\gamma_0 \phi_0^2} = \frac{2^{27} H}{m t_0^2 \log 2} = \frac{2^{27} H L^2 p_\star (d + p_\star)}{m \theta^2 b_0^2 \log 2}.$$

For $j \geq 1$,

$$\frac{1}{\gamma_j \phi_j^2} = \frac{2^{30} H}{m t_j^2 p_j} = \frac{2^{30} H L^2 (d + p_j)}{m \theta^2 b_0^2}.$$

It remains to show

$$p_\star (d + p_\star) + \sum_{j=1}^J (d + p_j) \leq C p_\star (d + p_\star)$$

for a universal C . Because $p_\star < d$, one has $J \geq 1$. Minimality of J gives $4^{J-1} p_\star < d$, and hence

$$J < 1 + \log_4 \frac{d}{p_\star} \leq 1 + \log_4 d.$$

The geometric-series formula and (11) give

$$\sum_{j=1}^J p_j = p_\star \sum_{j=1}^J 4^j = \frac{4p_J - 4p_\star}{3} < \frac{16d}{3}.$$

Moreover, $p_\star \geq A_0[1 + \log(d+1)]$, $A_0 \geq 1$, and $p_\star \geq 1$ imply

$$1 + \log_4 d \leq 1 + \frac{p_\star}{A_0 \log 4} \leq \left(1 + \frac{1}{\log 4}\right) p_\star.$$

Therefore

$$\sum_{j=1}^J (d + p_j) = Jd + \sum_{j=1}^J p_j \leq C p_\star d$$

for some universal constant $C \in (0, \infty)$. The desired inequality then follows by enlarging C . $\square$

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